Foundations of Large-Scale Multimedia Information Management and Retrieval

Lecture #5 Multimodal Fusion

Edward Y. Chang

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Foundations of Large-Scale Multimedia Information Management and Retrieval
Mathematics of Perception

Foundations of Large-Scale Multimedia Information Management and Retrieval *Mathematics of Perception* covers knowledge representation and semantic analysis of multimedia data and scalability in signal extraction, data mining, and indexing. The book is divided into two parts: Part I - Knowledge Representation and Semantic Analysis focuses on the key components of mathematics of perception as it applies to data management and retrieval. These include feature selection/reduction, knowledge representation, semantic analysis, distance function formulation for measuring similarity, and multimodal fusion. Part II - Scalability Issues presents indexing and distributed methods for scaling up these components for high-dimensional data and Web-scale datasets. The book presents some real-world applications and remarks on future research and development directions.

The book is designed for researchers, graduate students, and practitioners in the fields of Computer Vision, Machine Learning, Large-scale Data Mining, Database, and Multimedia Information Retrieval.

Dr. Edward Y. Chang was a professor at the Department of Electrical & Computer Engineering, University of California at Santa Barbara, before he joined Google as a research director in 2006. Dr. Edward Y. Chang received his M.S. degree in Computer Science and Ph.D degree in Electrical Engineering, both from Stanford University.

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Mathematics of Perception

TSINGHUA UNIVERSITY PRESS

Springer

In 1-2 Years (2004)…



+



=



Mobile devices with high-quality lens can record location, time, camera parameters, and voice

Image Organization & Retrieval

- Keyword-based
  - Manual labeling is subjective, cumbersome
  - The aliasing problem
- Content-based
  - Promising for general semantics: outdoor, landscape, flowers, people, etc.
  - Not enough for wh-queries (where, who, when, or what queries)

Image Organization & Retrieval

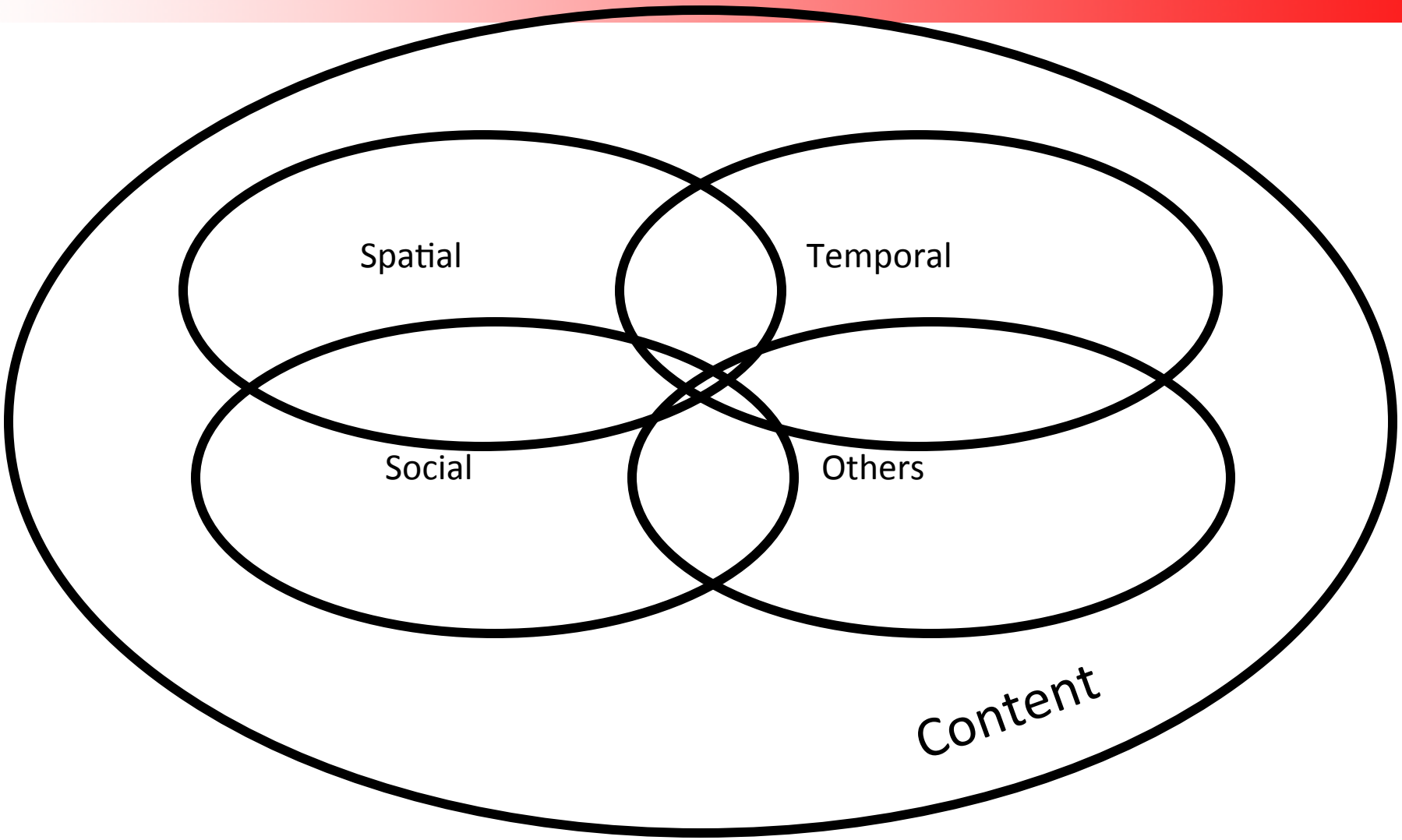
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KEYWORD + CONTENT

EXTENT = cont**EXT** + cont**ENT**

- Context
  - Spatial
  - Temporal
  - Social
  - Others (camera parameters and voice)
- Content
  - Perceptual features
  - Holistic features and local features

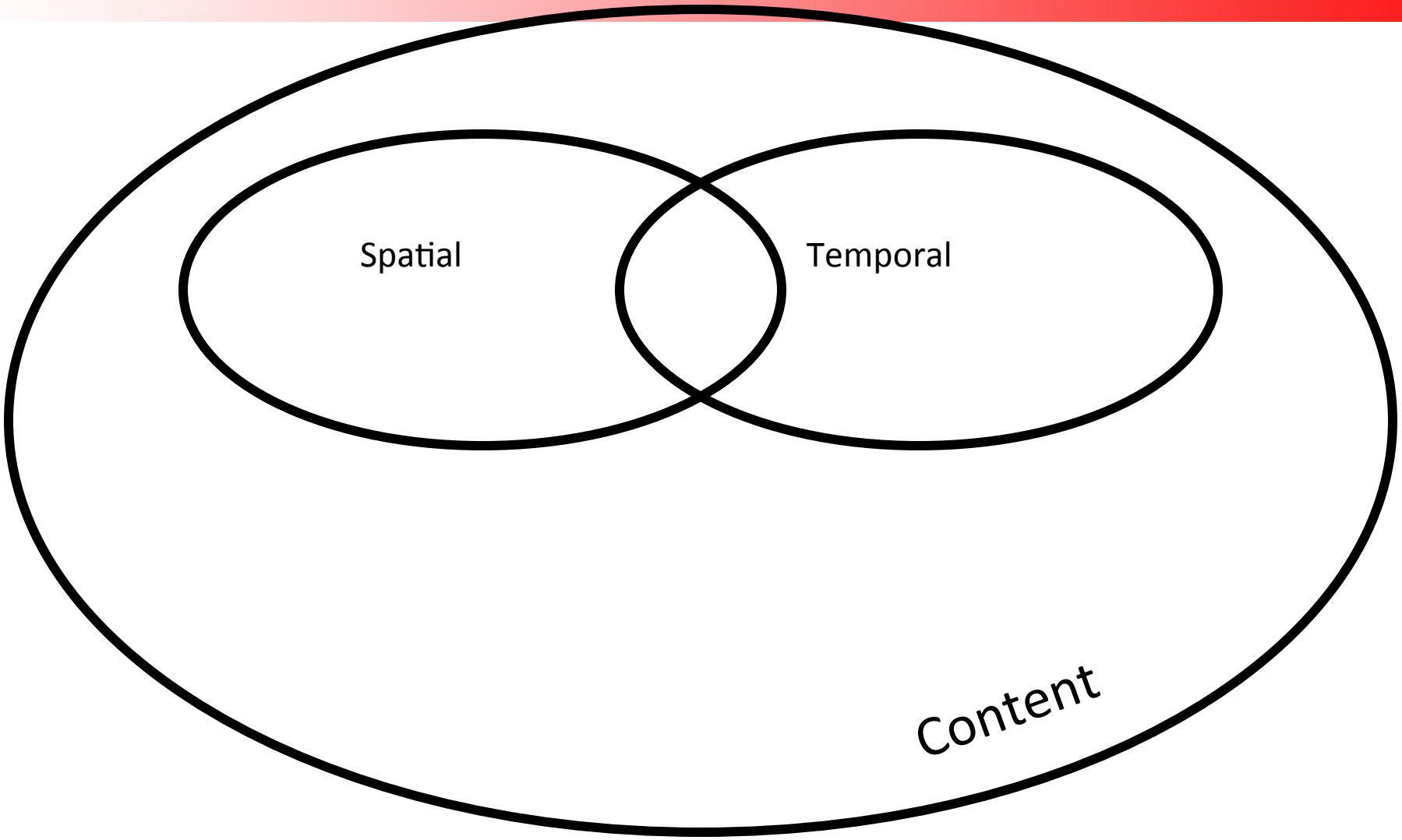
EXTENT



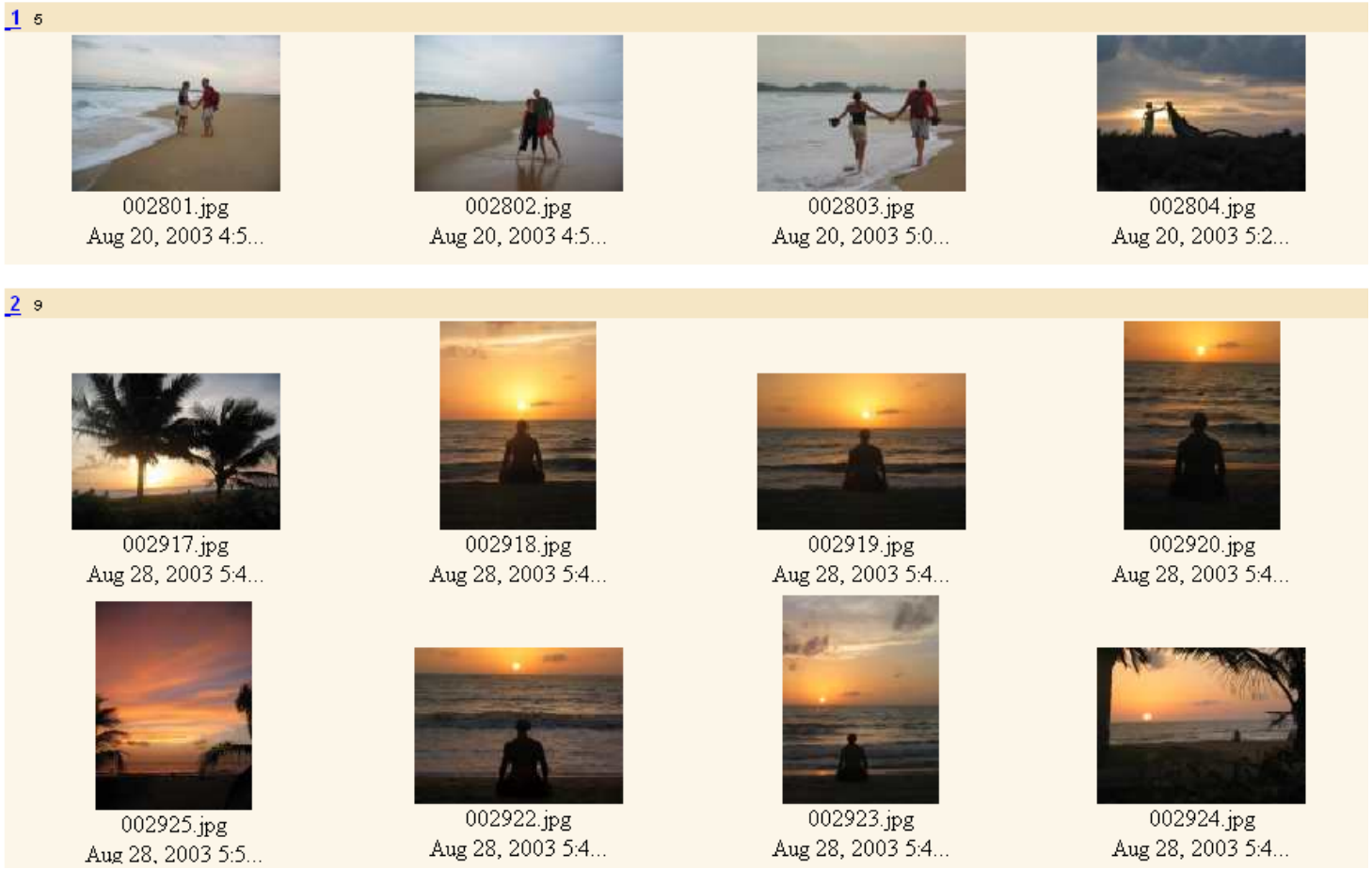
Context from Space/Time

- GPS or CellID data
  - into place names
- Time-based grouping
  - Into meaningful "events"
- From place names and time
  - Time of day
  - Weather (obtained from weather channel)

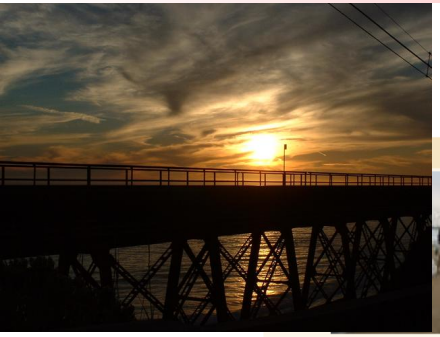
Example of Using Three Pieces of Information

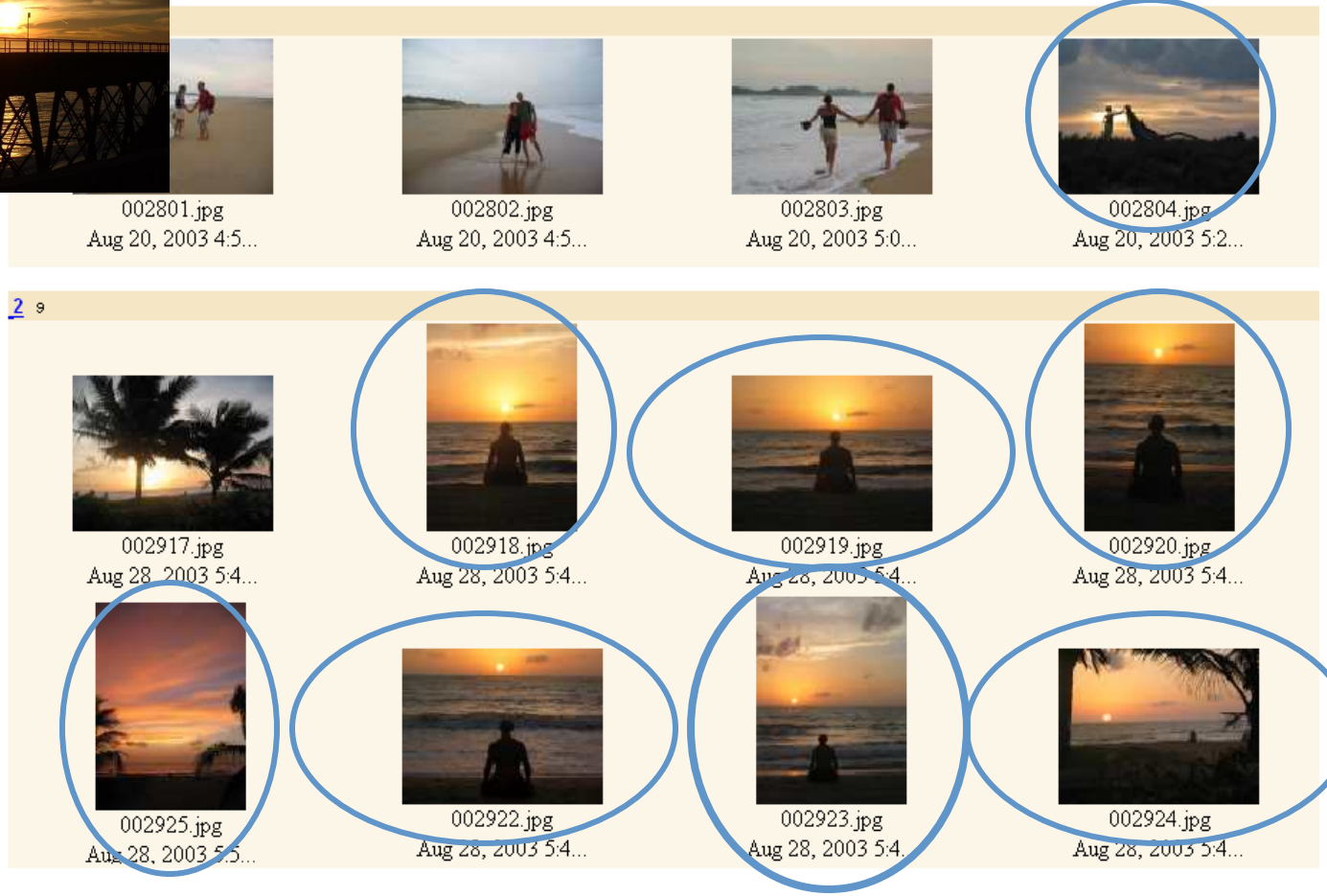


Maui Sunsets
can be obtained from Space/Time

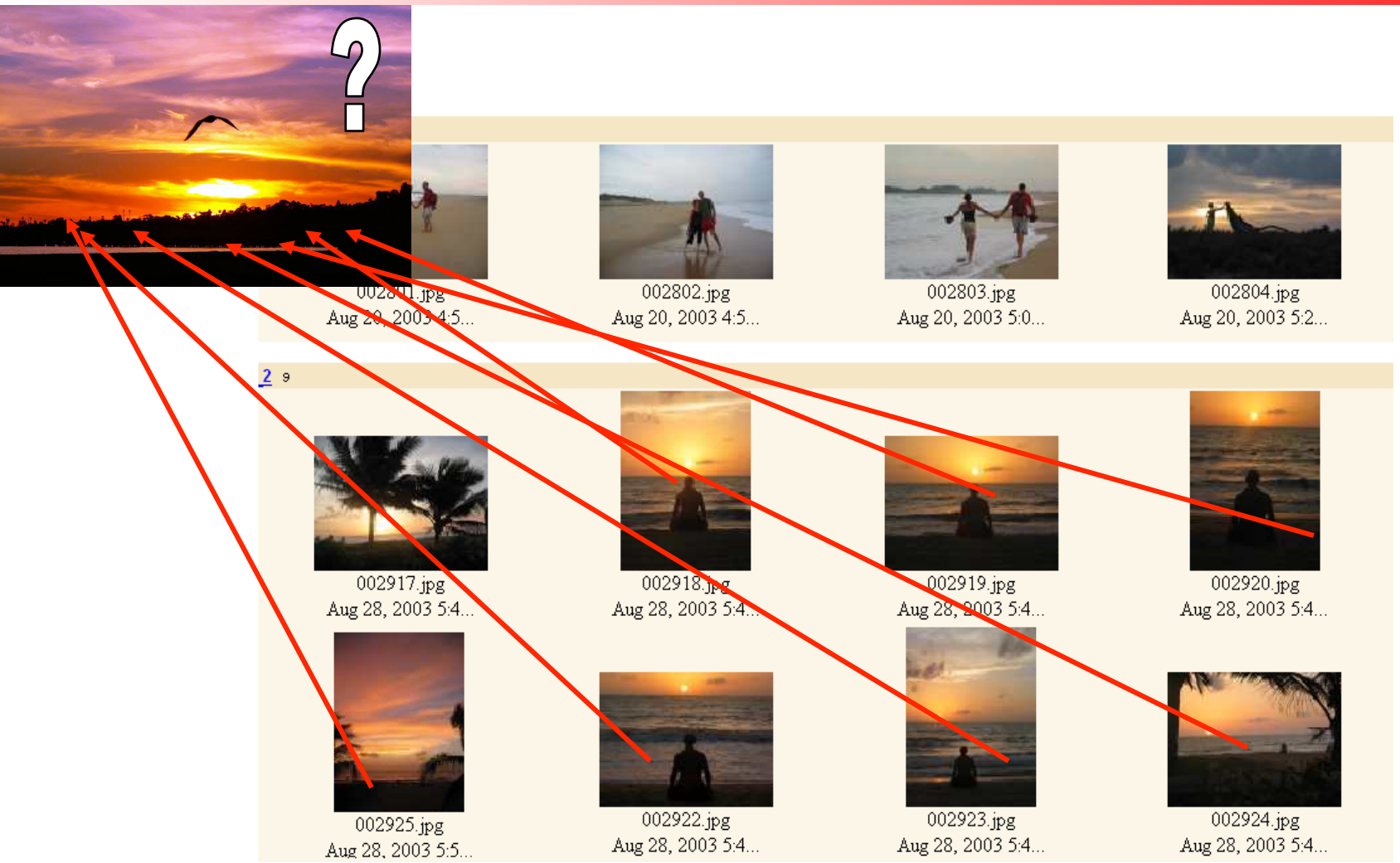


Use content for verification





Use content to transfer metadata



Summarize of the example

- Derived from Context
  - Derive time of the day
  - *Obtain location*
  - *Obtain camera orientation*
  - Obtain weather
  - Verify content
- Use of Content
  - Verify context
  - Transfer context

Other Key Semantics

- When (season)?
- Where (location)?
  - Landmarks
  - Indoor/outdoor
- Who (# of people, identity)?
- What?
  - Event

Where?

- A Golden Gate bridge photo can be taken at several places
- At a location, one can take photos of different landmarks, front/side, interior/exterior of a landmark

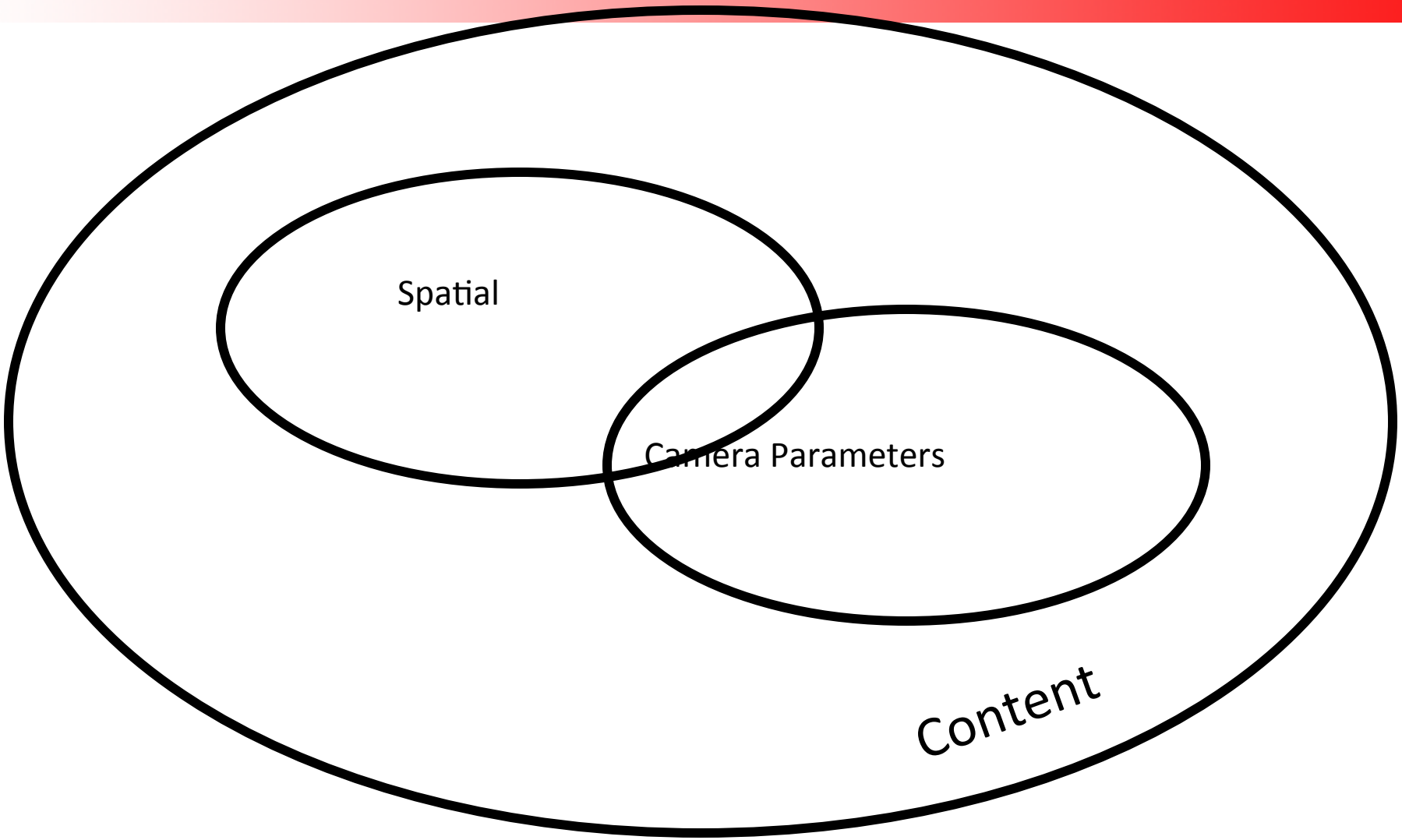


- A landmark may be occluded by other objects (cars, people)

What Landmark?

- Landmark identification is hard
- But with Geo information, EXTENT can
  - Limit the number of candidates by location and camera orientation
  - Use more robust content analysis methods
- Therefore, context + content is promising

Indoor/Outdoor?



Camera Parameters

- EXIF Metadata
  - Brightness
  - Flesh
  - Shuttle speed
  - Aperture
  - Focus length
  - Resolution
- GPS, Wifi, and INS signals

Who?

- Face recognition is a difficult
- Challenging issues are
  - Viewing angle, size, photo quality
  - Aging of a person
  - Occlusion
- But with context, we can
  - Limit the number of candidates (e.g., your social networks)
  - Use expensive and robust algorithms

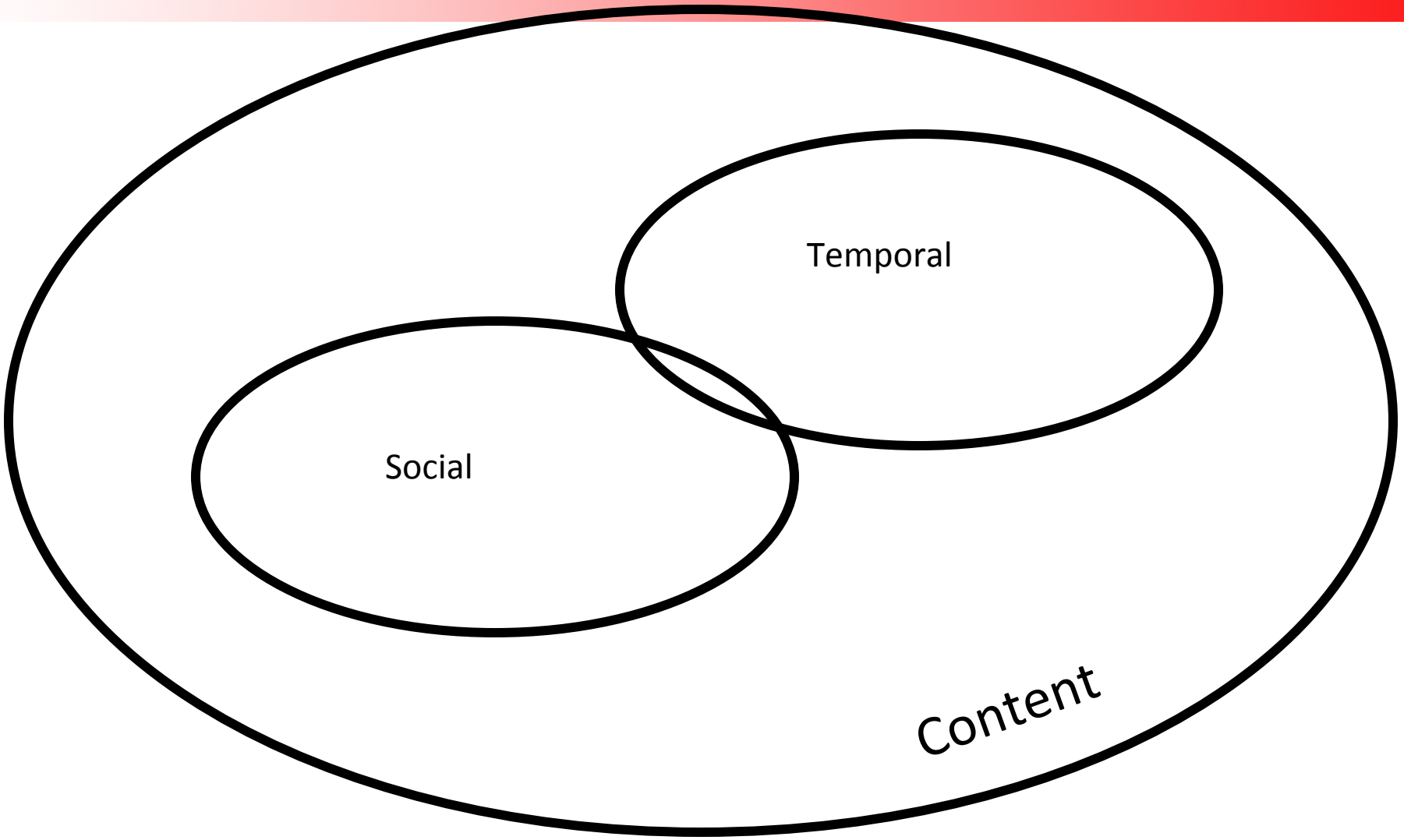
Who?

- What else can be helpful?





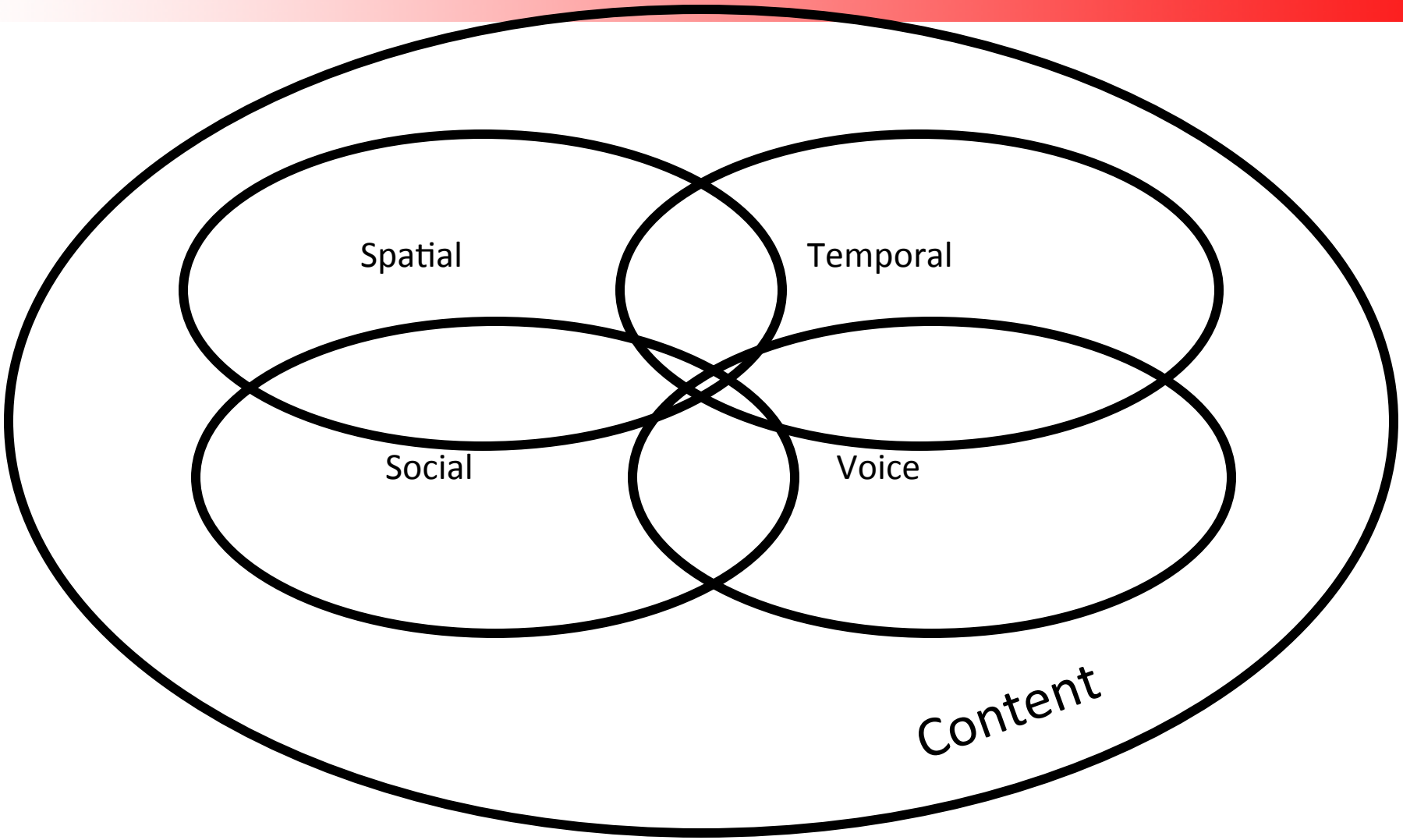
Who?



Grand Challenge: What Event?

- Birthday
- Graduation
- Wedding
- Vacation
- etc.

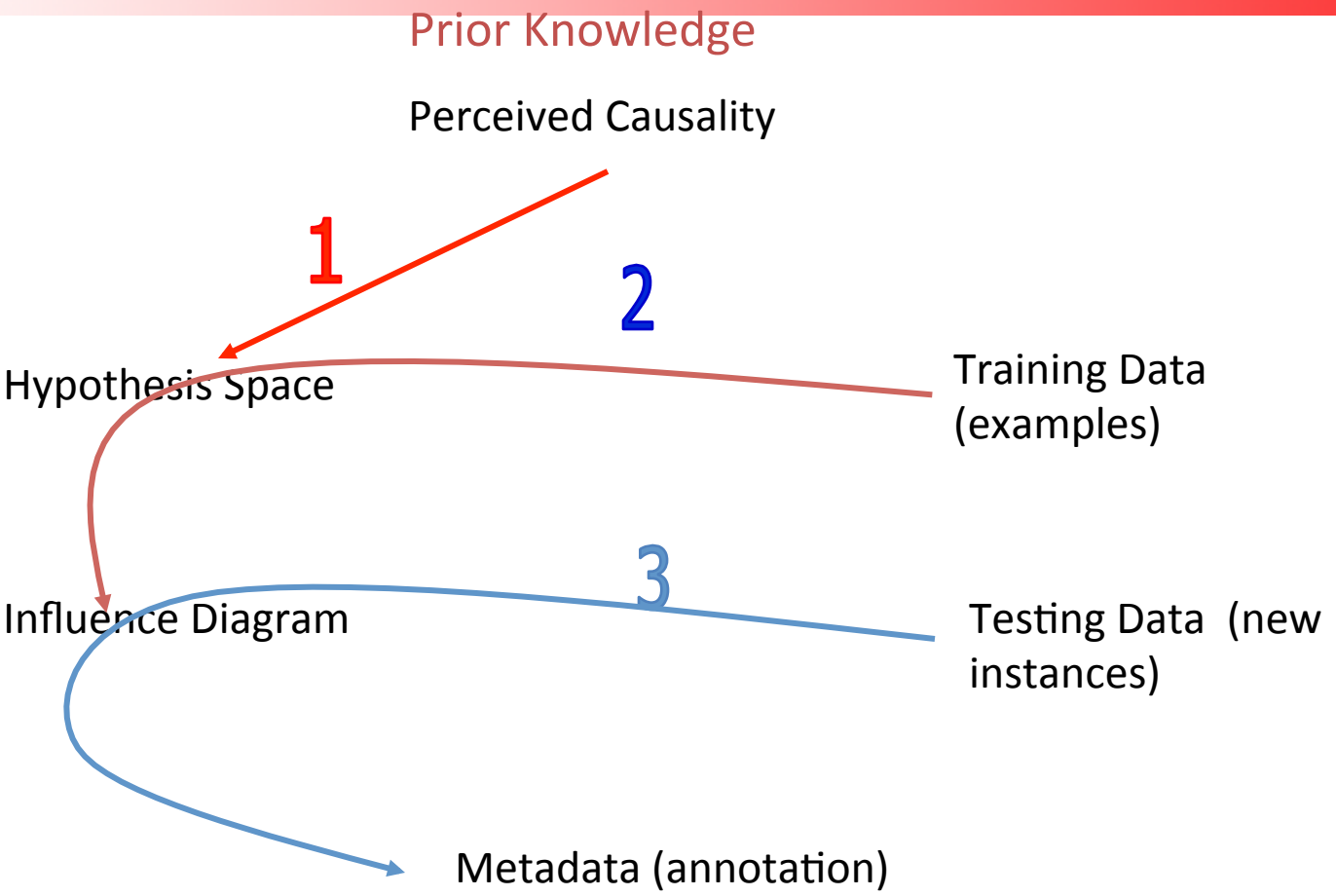
What Event?



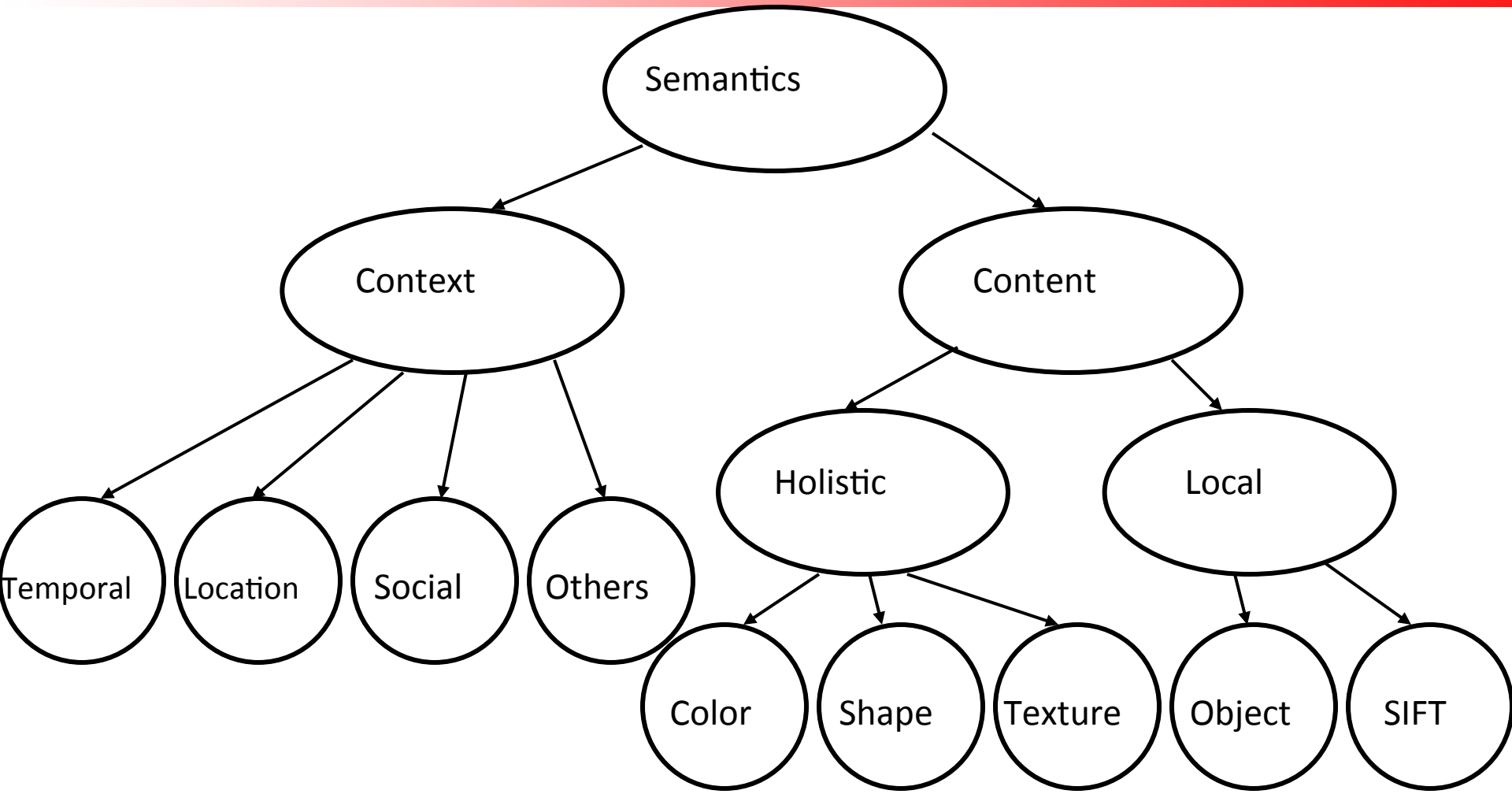
EXTENT

	When	Where (places, landmarks)	Where (Indoor/ outdoor)	Who	What (event)
Spatial		○	○		○
Temporal	○	○		○	○
Social				○	○
Exif/Voice			○		○
Content	○	○	○	○	○

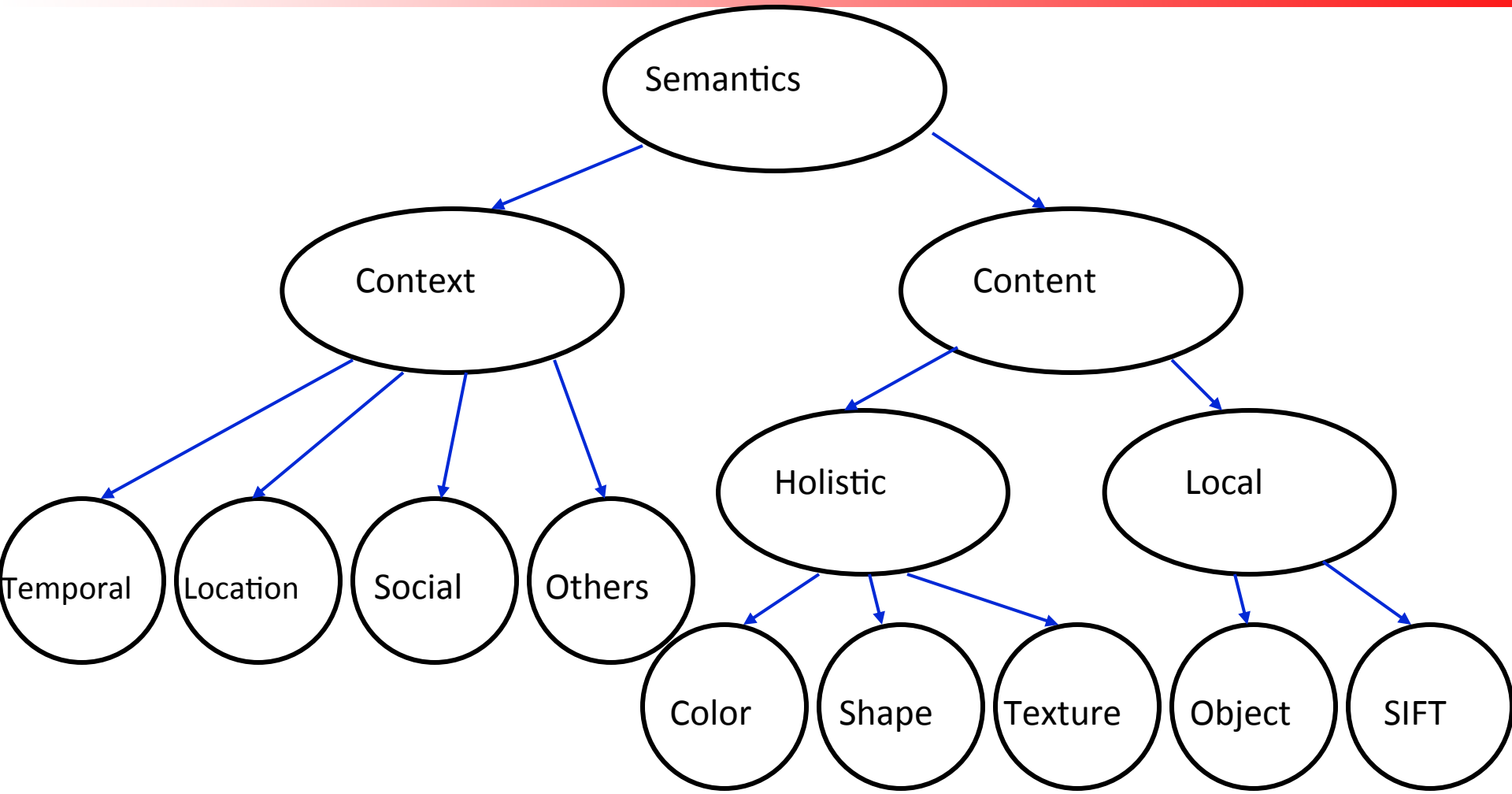
EXTENT Fusion Model



Variables and Structures



Conditional Probabilities

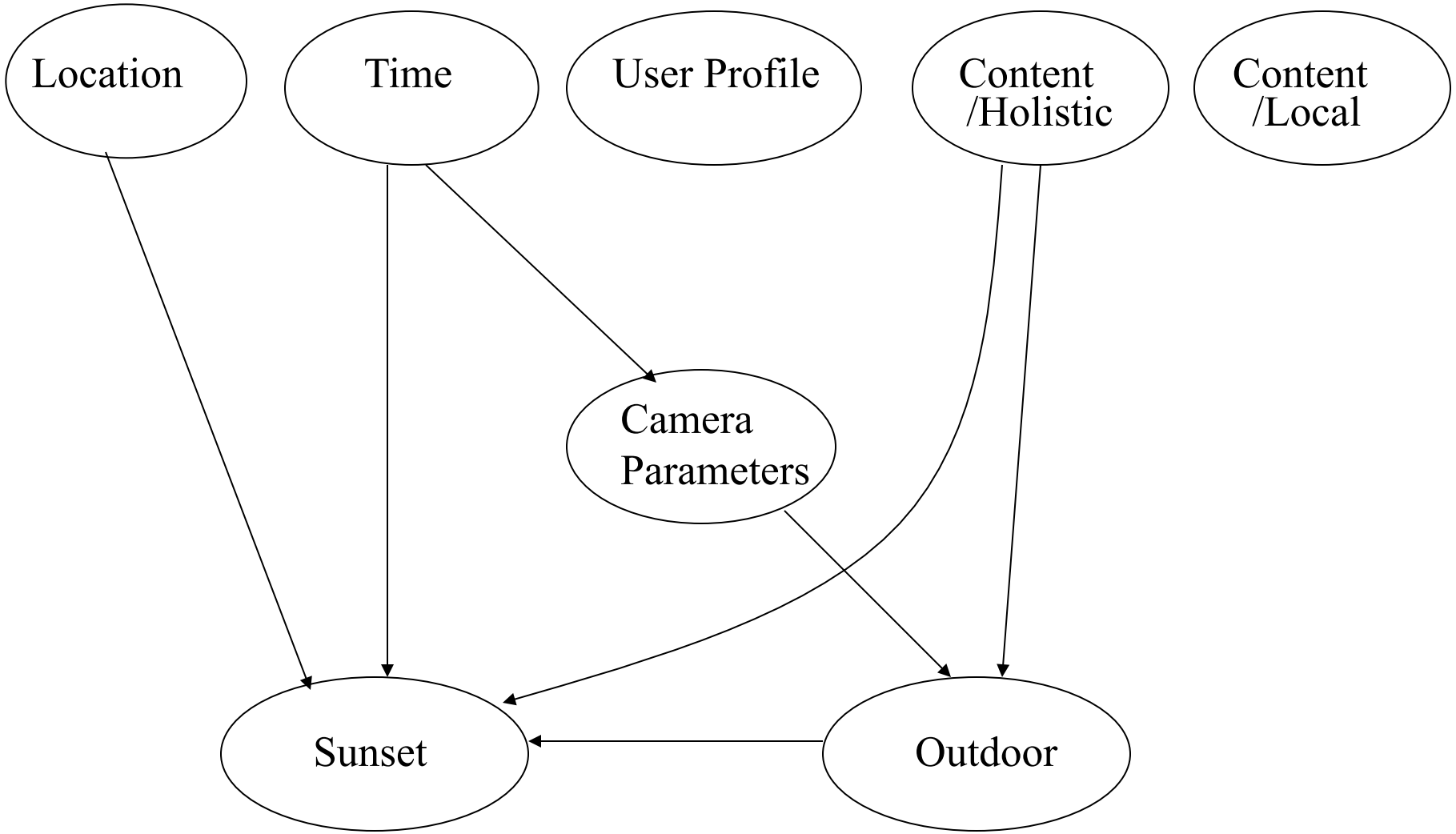


“Beach” Case Study

- More Probable
  - Outdoor
  - Day time
  - With ocean (blue)
  - Beach party
  - Without people?
- Less Probable
  - Indoor
  - Night time
  - Without ocean
  - With many people?

"Beach" Case Study

- More Probable
  - Outdoor
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  - Beach party
  - Without people?
- Less Probable
  - Indoor
  - Night time
  - Without ocean
  - With many people?


Location
Time
User Profile
Content /Holistic
Content /Local
Camera Parameters
Sunset
Outdoor

From OPON to IPIN

outdoor position outdoor navigation

to *indoor position indoor navigation and location's help to multimedia research*

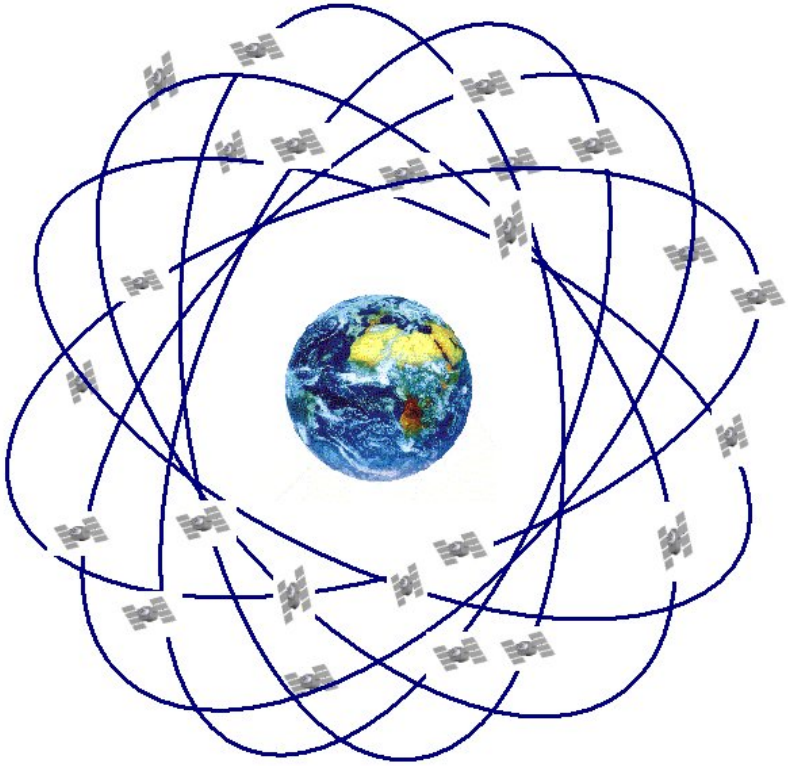
Outline

- Existing Technologies
  - GPS, Cell Tower, WIFI
- Shortcomings
  - Signal availability
  - Power consumption
- XINS
  - X
  - INS
  - X + INS Fusion

GPS

Outdoor --- GPS

- Global Positioning System (GPS)
- Since 1973, fully operational in 1994
- 24 satellites divided into 6 groups of 4
- Each satellite circles the earth on a precise orbit
- Each group constructs an orbit plane
- A receiver accepts signals from multiple satellites, then can determine its latitude, longitude, and height

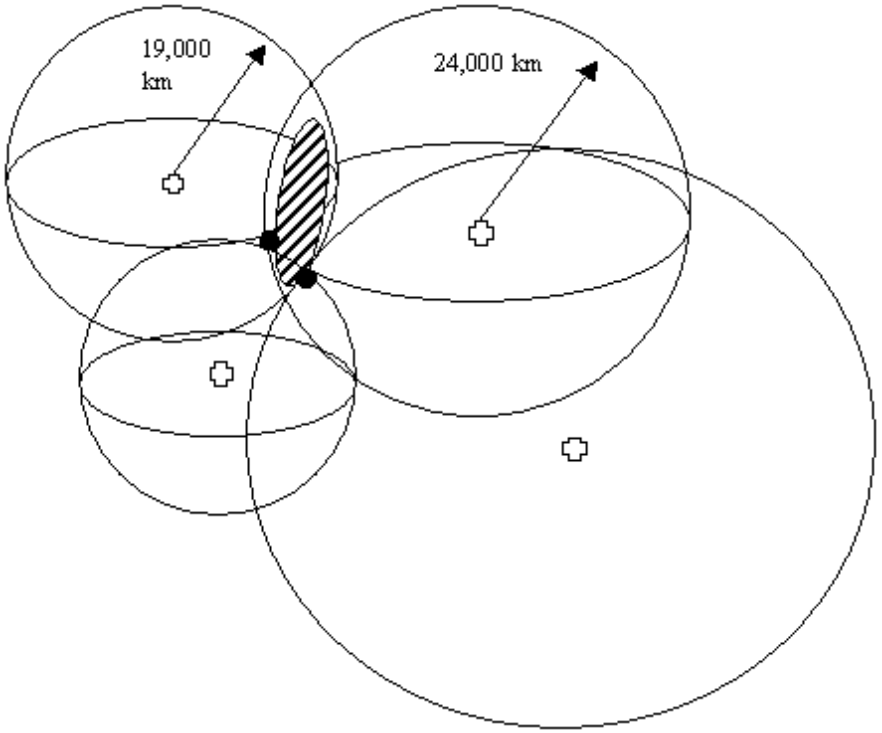


Basic Theory

- Based on trilateration
- To trilaterate, GPS measures distance from at least four satellites
- Measure distance requires good clocks
- Obtain satellite positions
- Differential correction

How many satellites do you need?

- One narrows the possible positions to a sphere
- Two help locate you on a circle
- Three limit the possibilities to 2 points
- Four may confirm the final location.
- The more, the better



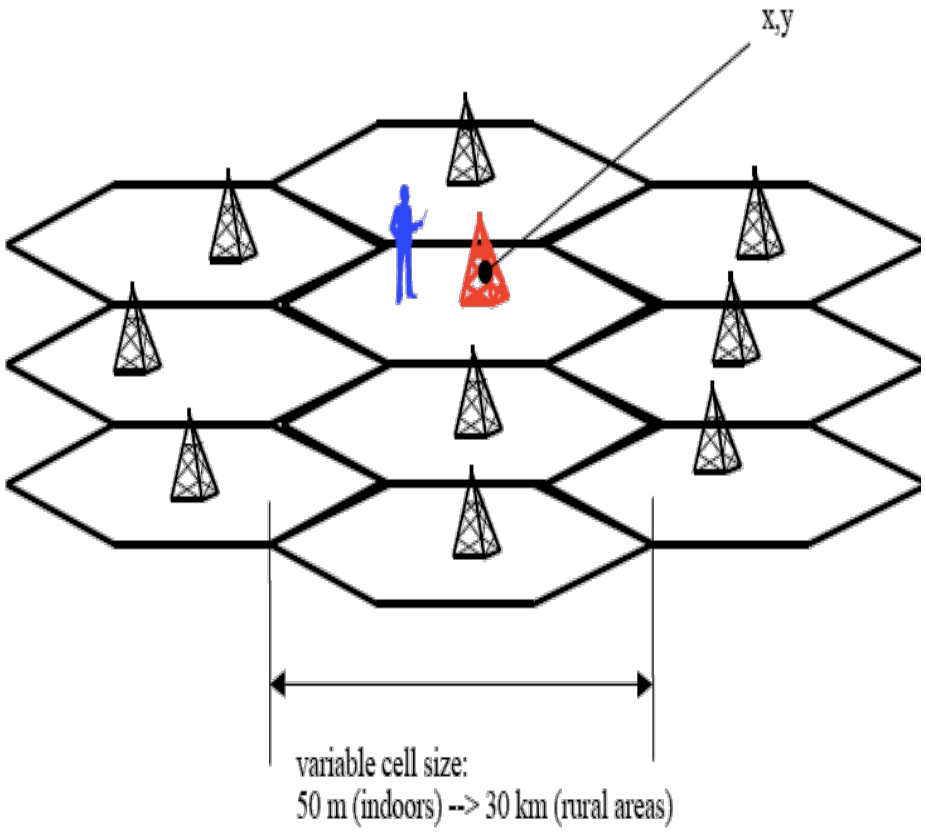
GPS Shortcomings

- TTFF (Time to First Fix) too long, usually > 30s
  - AGPS, Assisted GPS is based upon providing GPS satellite / location information to the handset, via the data network
- Urban environment
- Indoor locations
- Power consumption

Cell ID

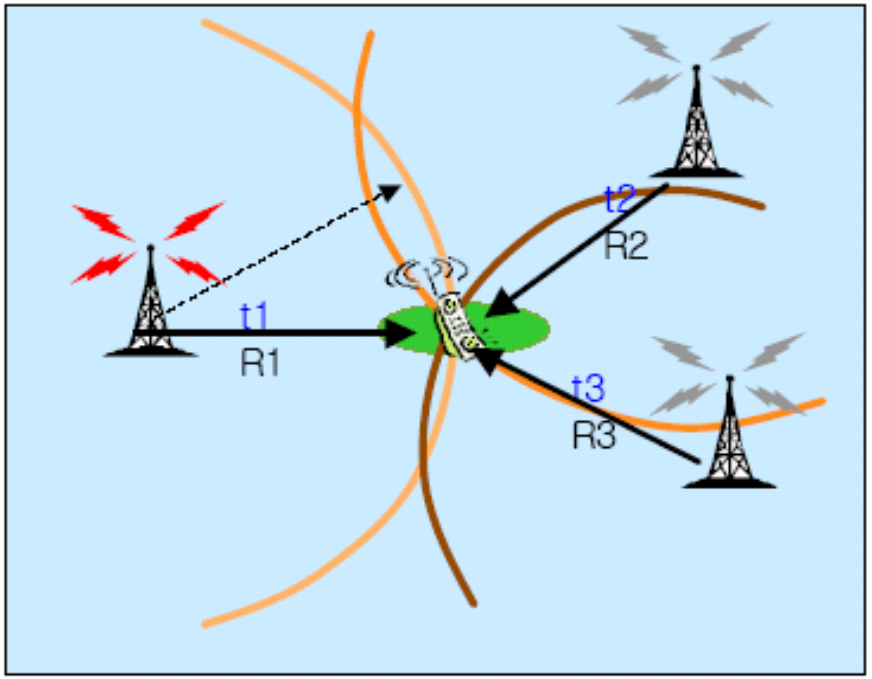
Basic Theory

- Based on the Cellular network infrastructure
- Mobile device generally connects the nearest cellular network tower
- Accuracy is dependent on cell density
- May be network-based or handset-based



Trilateration - TOA

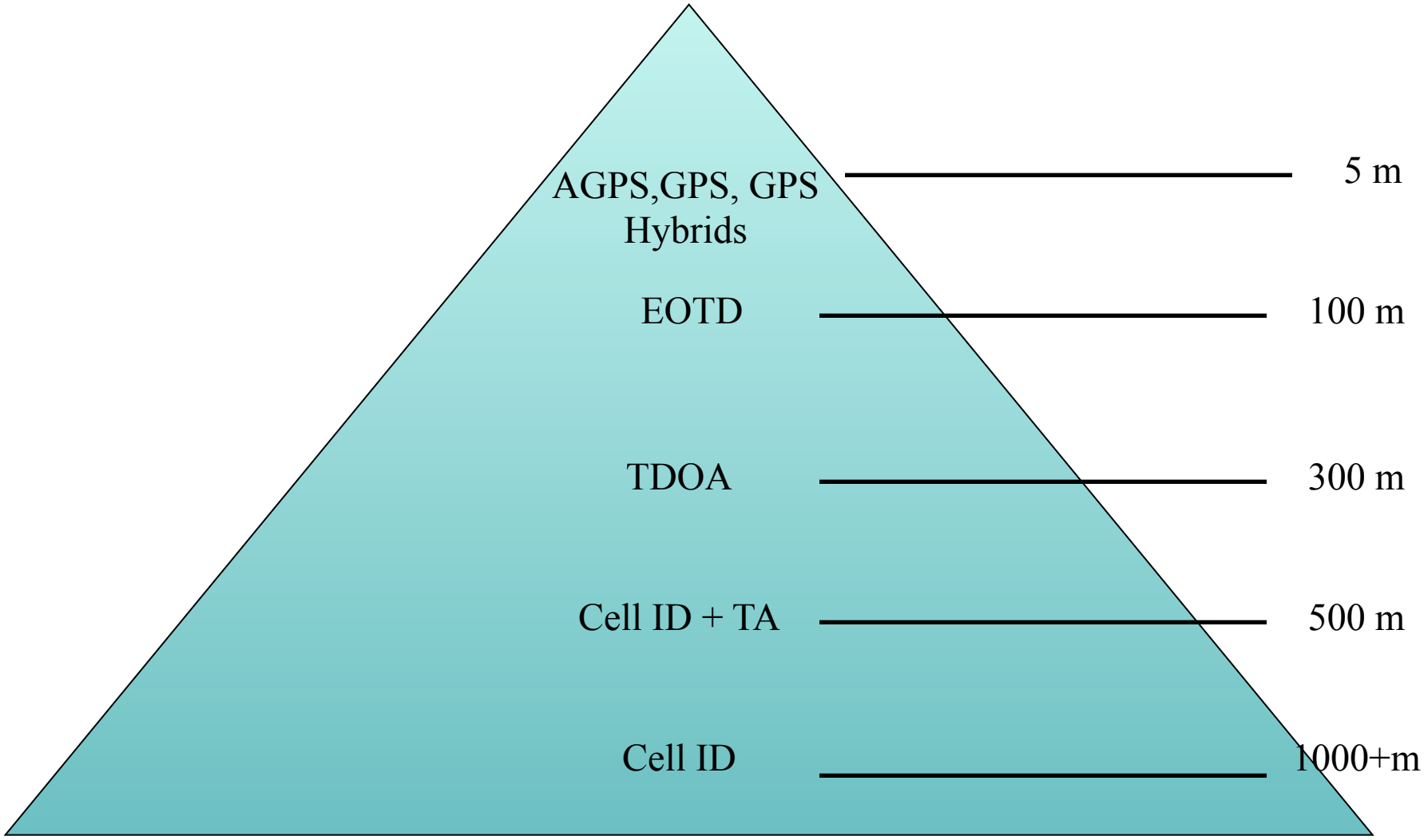
- Measure the Time of Arrival (TOA) of signals between the cellular tower and mobile
- Mobile device locates at the intersection point which will be made by more than 3 circles



More advanced and Hybrids

- CellID + Timing Advance (TA)
- Triangulation - AOA
- Multilateration – TDOA
- Enhanced Observed Time Difference (EOTD)
- EOTD + AGPS

Positioning Accuracy



Google Location Server (GLS)*

- Operators know where their towers, but Google does not… ☹
- Two-step, crowd-sourcing approach
  - Estimate cellular tower location using crowd-source data
  - At serving time, depends on the content in the query, compute device location using: nearest tower, neighbor towers, or fingerprint
- Advantages
  - Crowd sourcing does not required lengthy negotiation and contract with every service provider
  - Easier to support various devices
- Disadvantages
  - Crowd-source data can be full of noises
  - Keeping up with changes in network is still challenging

*Tsuwei Chen, "Location Determination for Mobile Devices", 2010, ShangHai

Wifi

Two Models

- Wave Propagation
  - Based on computing distance between a mobile device to known AP locations
  - Obstructions, deflection, noise can affect accuracy
- Signal Strength Map (SSM) or RF Fingerprint
  - Signal strengths at location map
  - Each location has a signature (fingerprint)
  - Measurements matching can be deterministic or probabilistic
  -

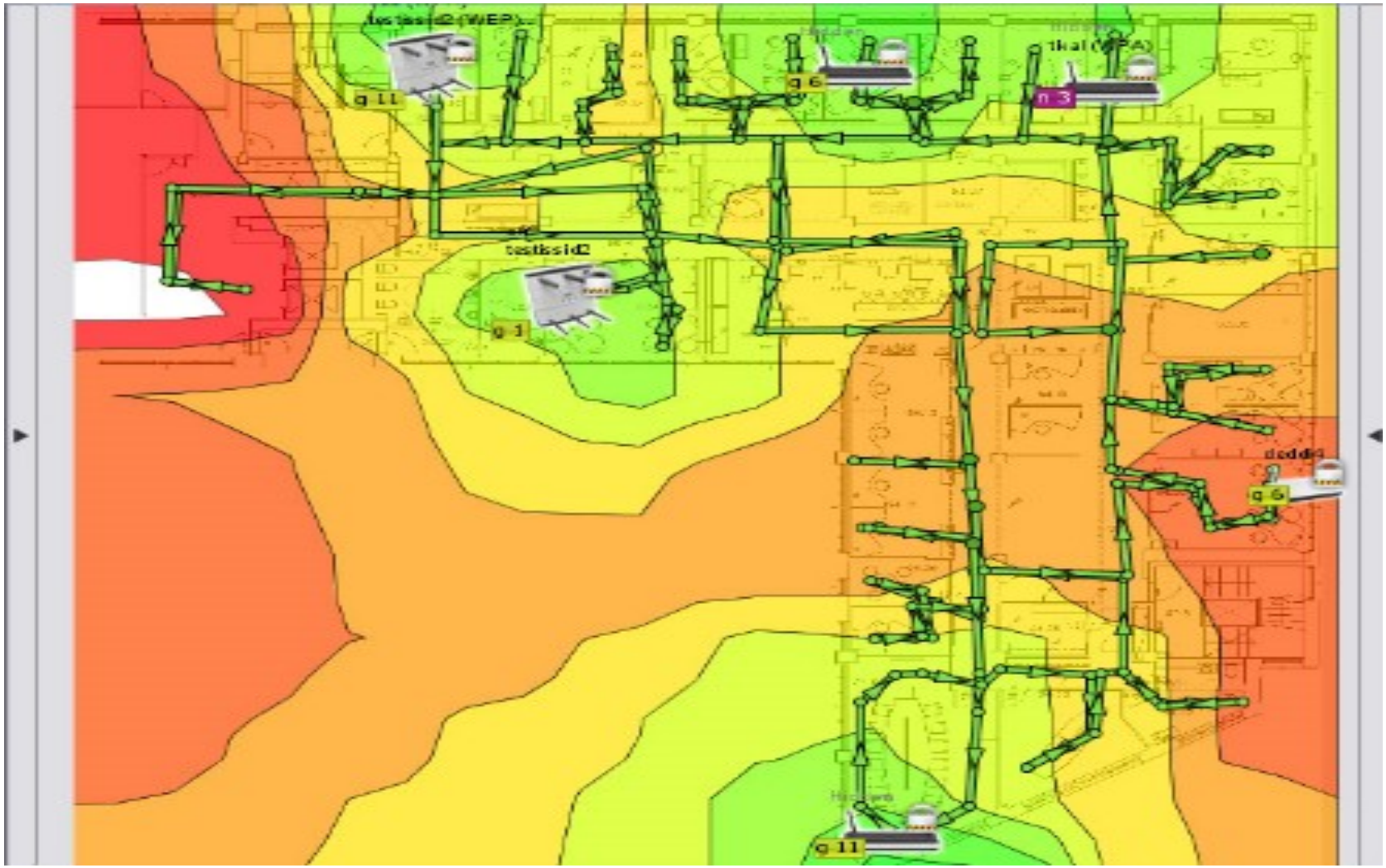
Wave Propagation Computation

- Polynomial model
  - Expressing distance according to signal strength
- Friis model
  - $\frac{P_R}{P_T} = G_R G_T (\frac{\lambda}{4\pi d})^k$
  - P_R : power receiver; P_T: power transmitter;
  - G_R : receiver antenna gain; G_T : transmitter antenna gain; λ: signal wavelength; *k*: parameter
  - *k*= 2 for atmosphere; *k* = 3.5 was shown good for some buildings

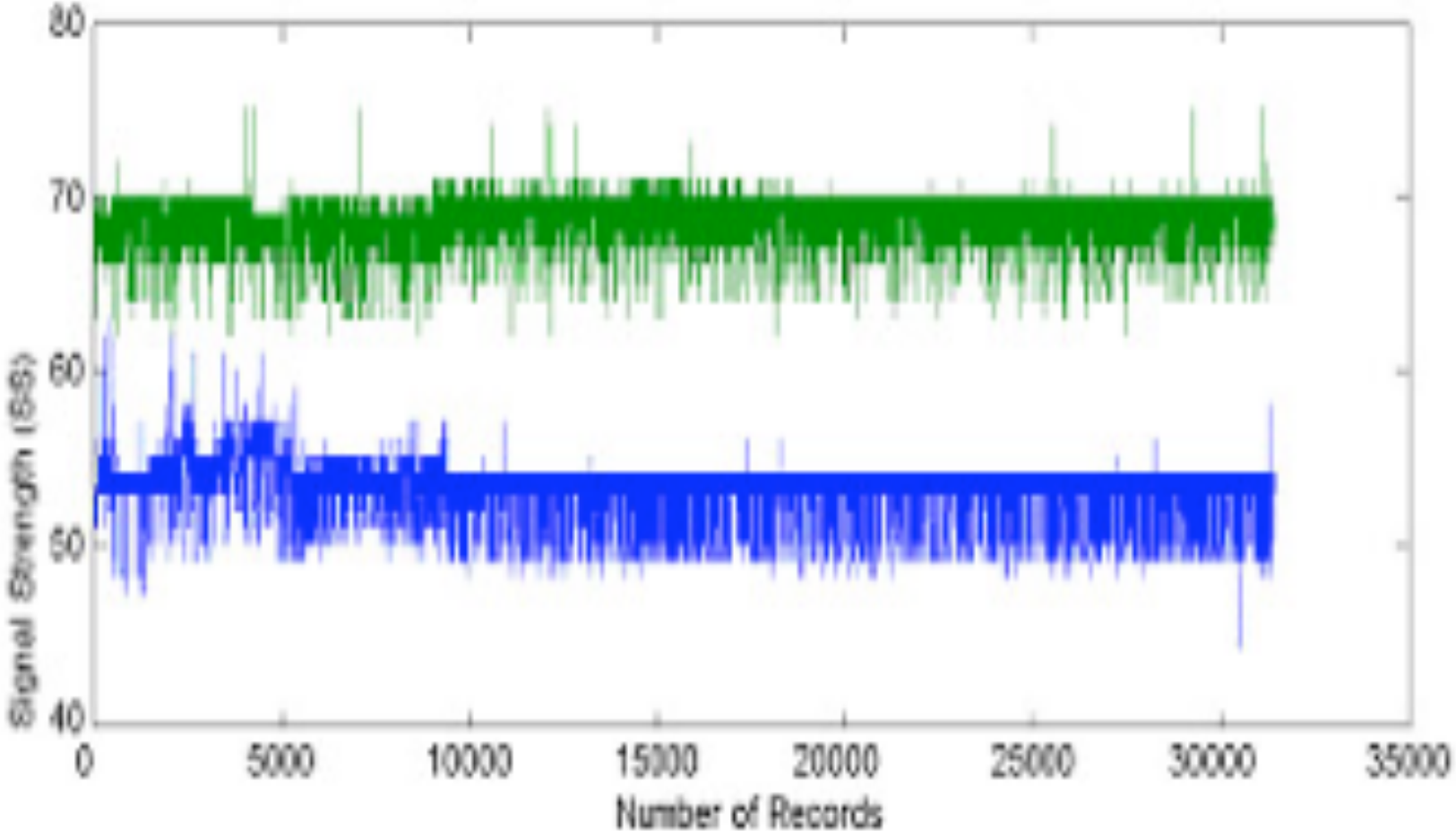
SSM (RF Fingerprint) 2x2

- Step #1 Build a signal strength map
  - By measurement, time consuming
  - By simulation, not considering physical environmental factors such as obstacles
  - Modeling transient noise (people, door, weak signals)
- Step #2 Online positioning
  - Matching deterministically or probabilistically
  - But multiple points may have the same "signature"
  - Viterbi algorithm may help by considering history
  - Floor map may help to constrain multiple neighboring points

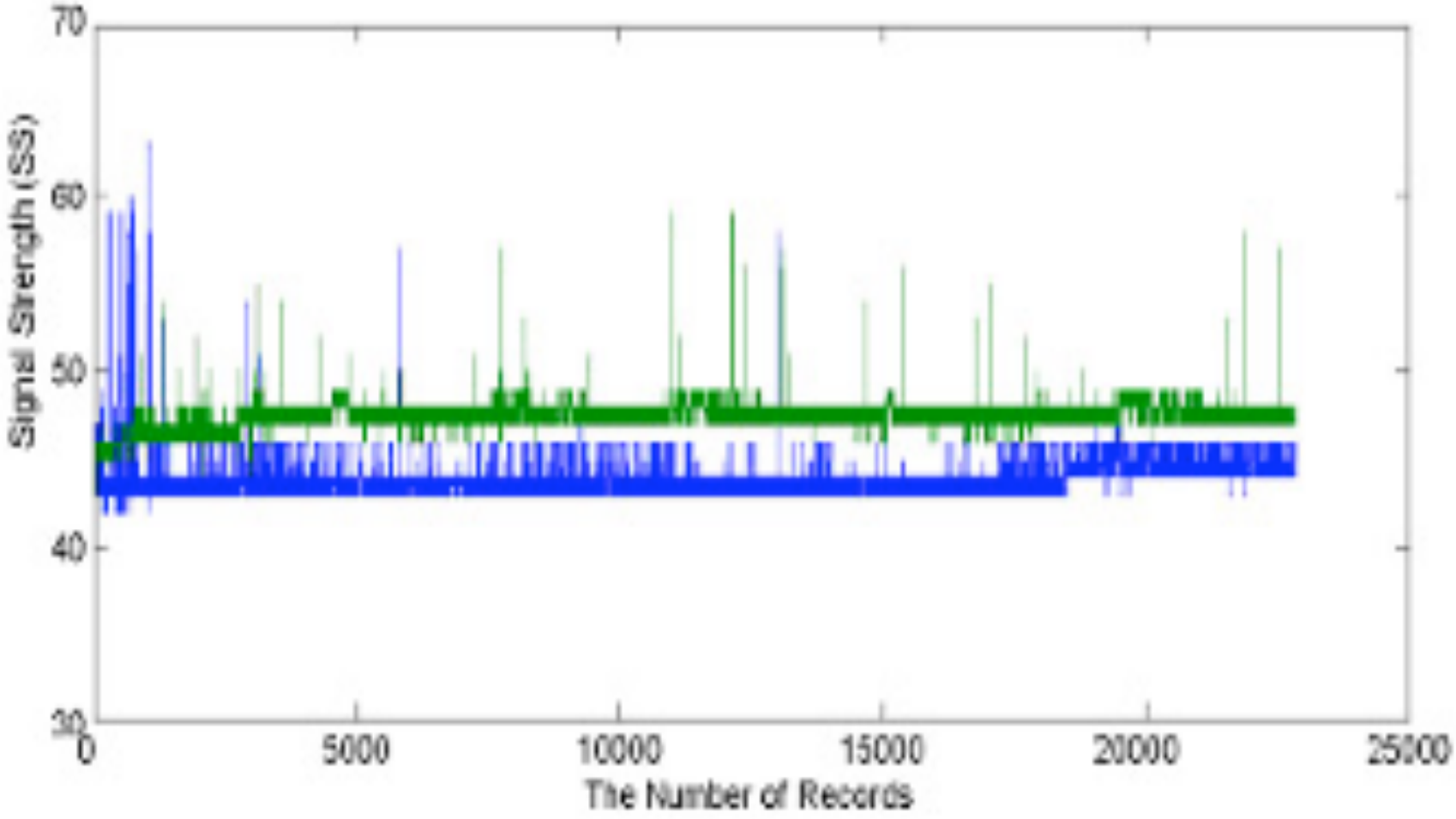
Heat Map (SSM)



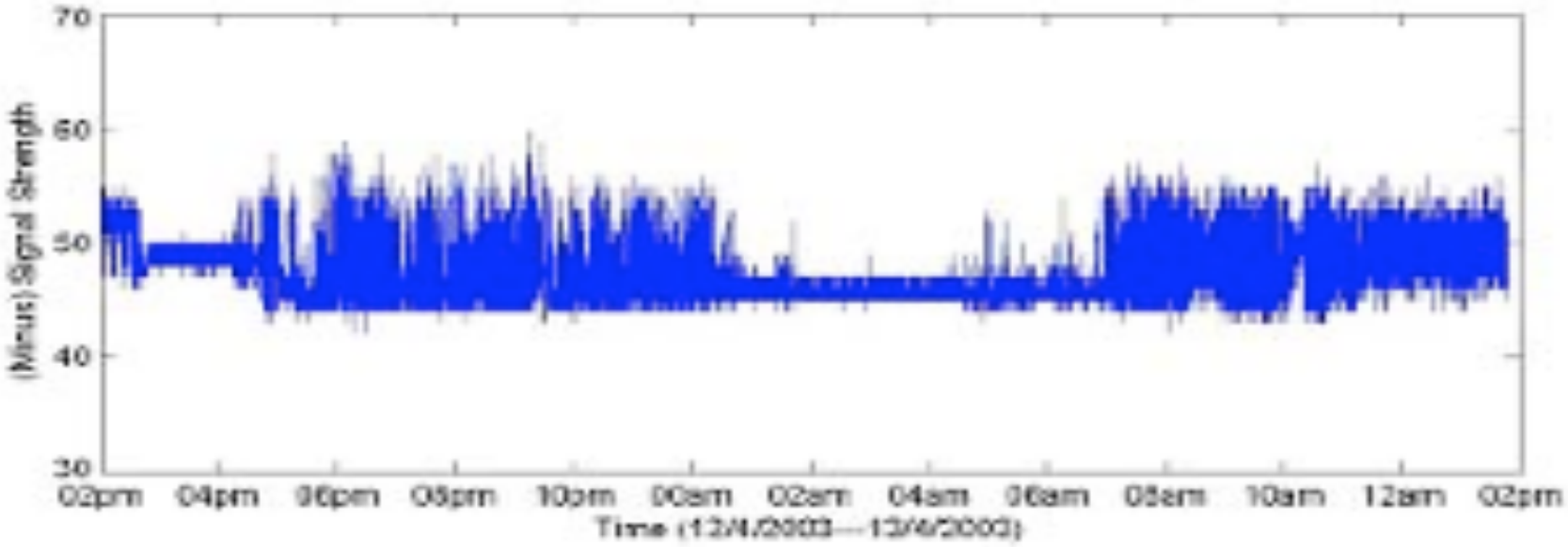
Wall Effect (Solid Wall)



Wall Effect (Cubicle Wall)



SS 24-hour Measurement (People Effect)



- Sampling rate ½ second
- Noise free reading between −45 to −47 dBm
- Noise ranges from 5 to 10 dBm

Propagation vs. SSM

	Accuracy	Setup
Propagation	5 – 10m	Easy
SSM	4m (about)	Time-Consuming

XINS

simple in concept, complicated in execution

What is XINS?

- *X* is a reliable initial condition, *X* can be
  - GPS positioning
  - Wifi positioning
  - P2P synchronization
- INS --- Inertial Navigation System
  - Consisting of accelerometers, gyroscopes, or other motion sensing devices
  - Used in guided missiles, aircraft, ships, spacecraft
  - Compute direction and speed of movement
  - With only infrequent external references

Analogy: Car's position in a tunnel?



INS

- Measure linear and angular acceleration, to estimate
  - Change in velocity
  - Change in position
  - Change in orientation
- Gyroscopes
  - Measuring angular changes
- Accelerometers
  - Measuring linear acceleration

Acceleration → Velocity → Position

- Gyroscopes
  - Get attitude readings, from which we can get moving directions
- Accelerometers
  - New position = initial position + displacement
  - Displacement = integrate on velocity over time
  - Velocity = integrate on acceleration over time

History

- WW II German guidance system: two gyroscopes with one accelerometer
- Later used in cruise missiles, aircraft, etc.
- At least one sensor for each direction: pitch (nose up or down), yaw (nose left or right), and roll (clockwise, counter-clockwise)

Technical Challenges

- Error on the INS devices
  - Zero drift
  - Precision bias (one full rotation does not read 360°)
- Obtaining good *initial*, *recalibrate* points
- Robust fusion models
  - Integration drift - Error is cumulative and increase in time because of integration

Zero Drift Correction

Prior Approaches

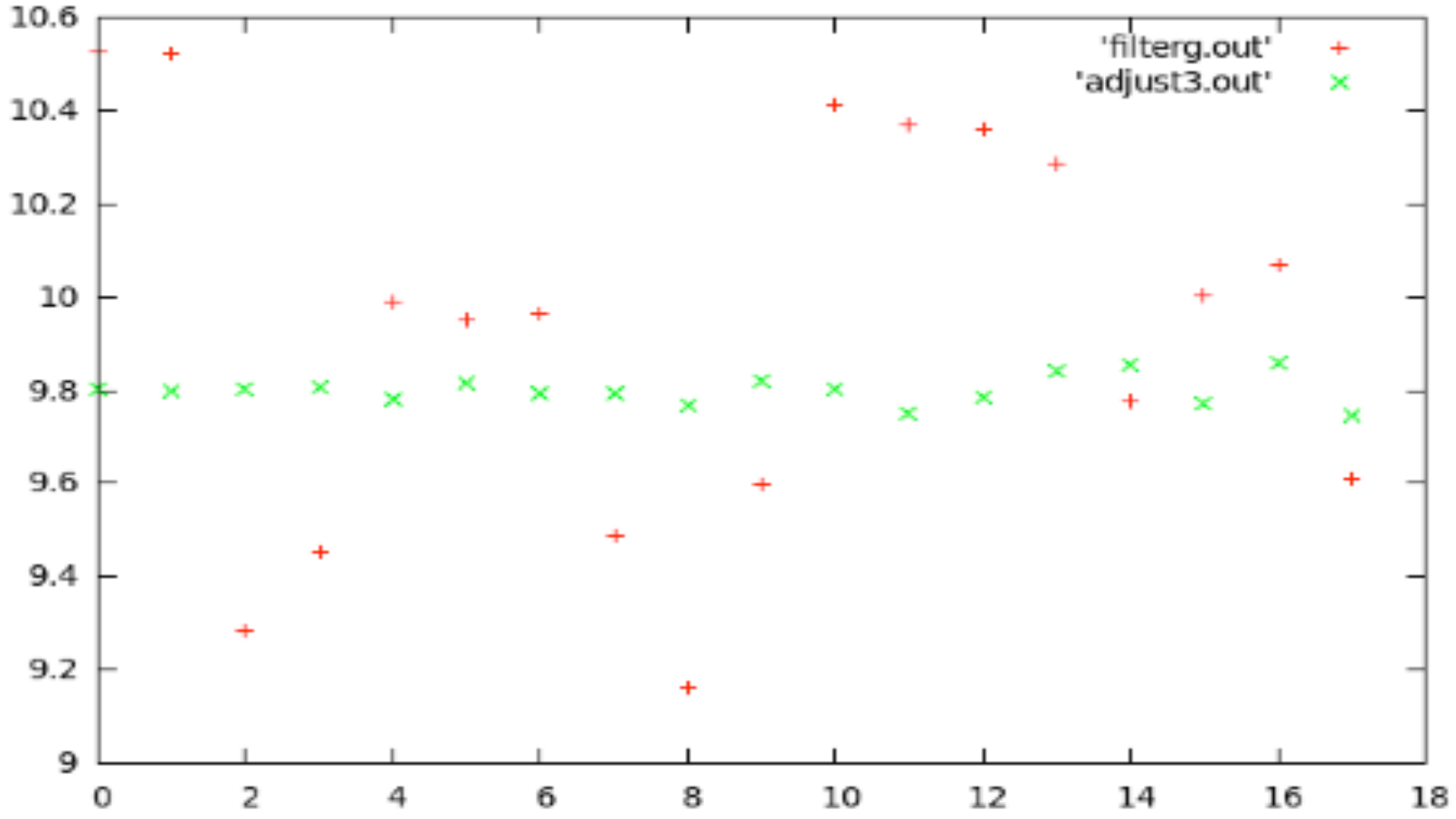
- Two gyroscopes configured at the opposite orientations, cancelling drifts
- Calibrate in the stationary state for 30s, every few hours

Downhill Simplex

- Main Idea
  - The collected uncalibrated data form an eclipse
  - Transform the eclipse into a circle
  - Yield six transformation parameters
  - Use the parameters to calibrate accelerometer
- Significant Advantage
  - No redundancy required
  - Simple procedure, can be done for each phone

Down-Hill Simplex

Adjusting parameters to fit reading to *g*

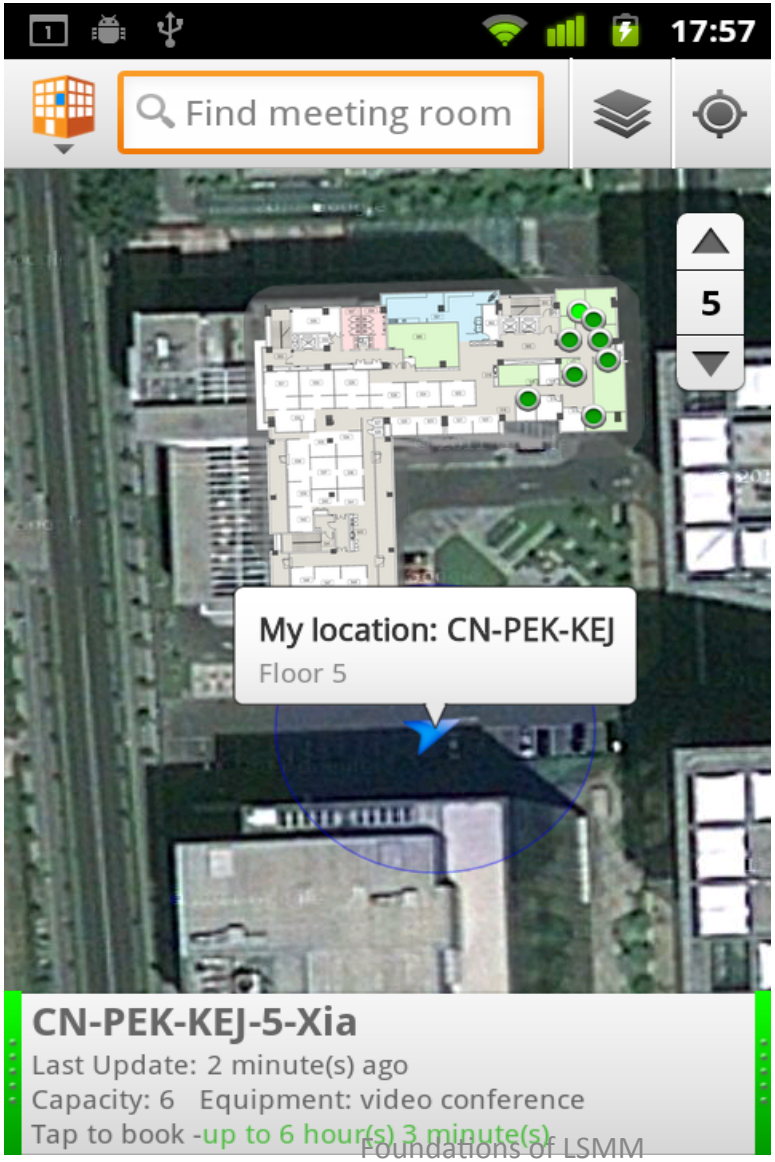


Signal Fusion Using Kalman-Family Filter

- Kalman Filter is using observations with noise to predict the actual value
- Initial conditions
  - GPS
  - AGPS
  - WIFI
  - P2P signals
- Derivatives
  - INS devices

Apps

Indoor locating



INS Assisted Panoramas

Using Inertial Navigation System (INS) to help image alignment and stitching for panorama on mobile phones



INS-Assisted Panoramas

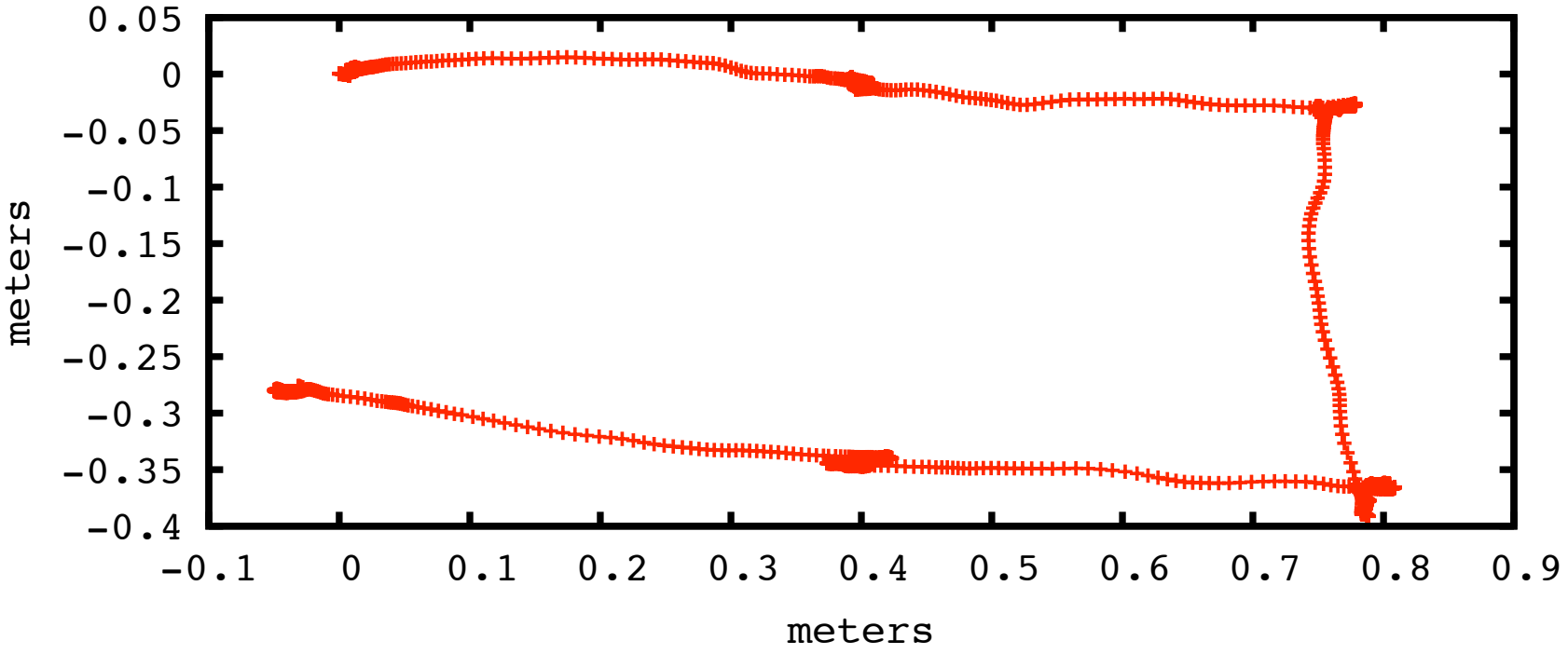







INS-Panoramas

Hand movement trajectory



INS-Panoramas

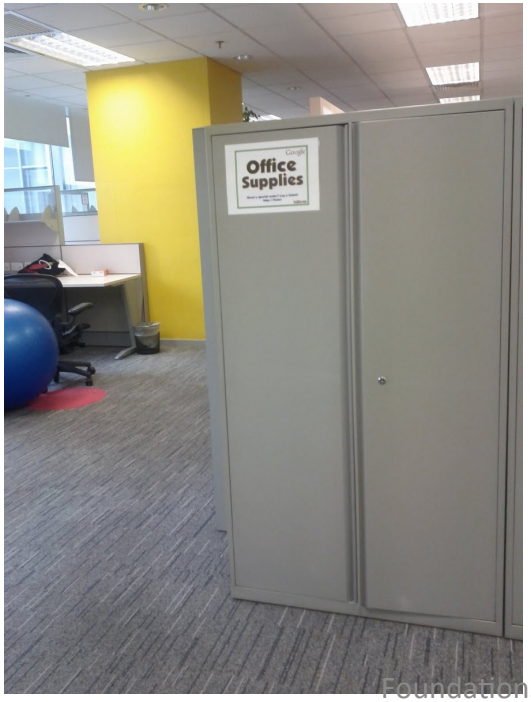
INS assisted stitching, faster and better



Quality Enhancement

- Homo-structure







Quality Enhancement

Without INS

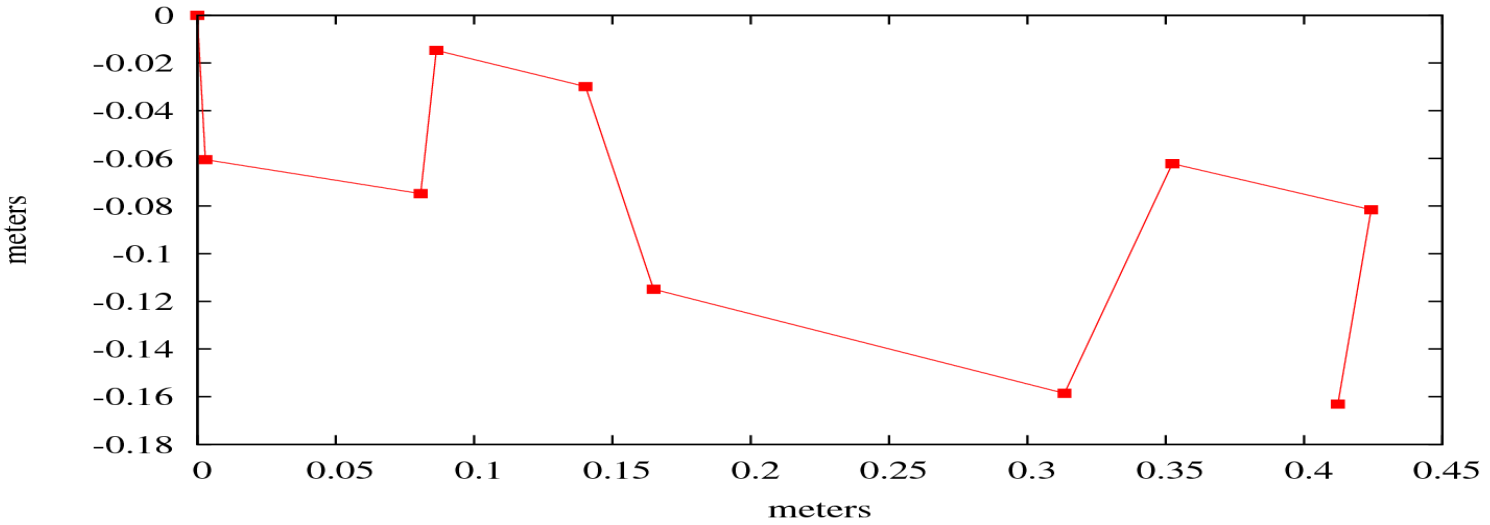


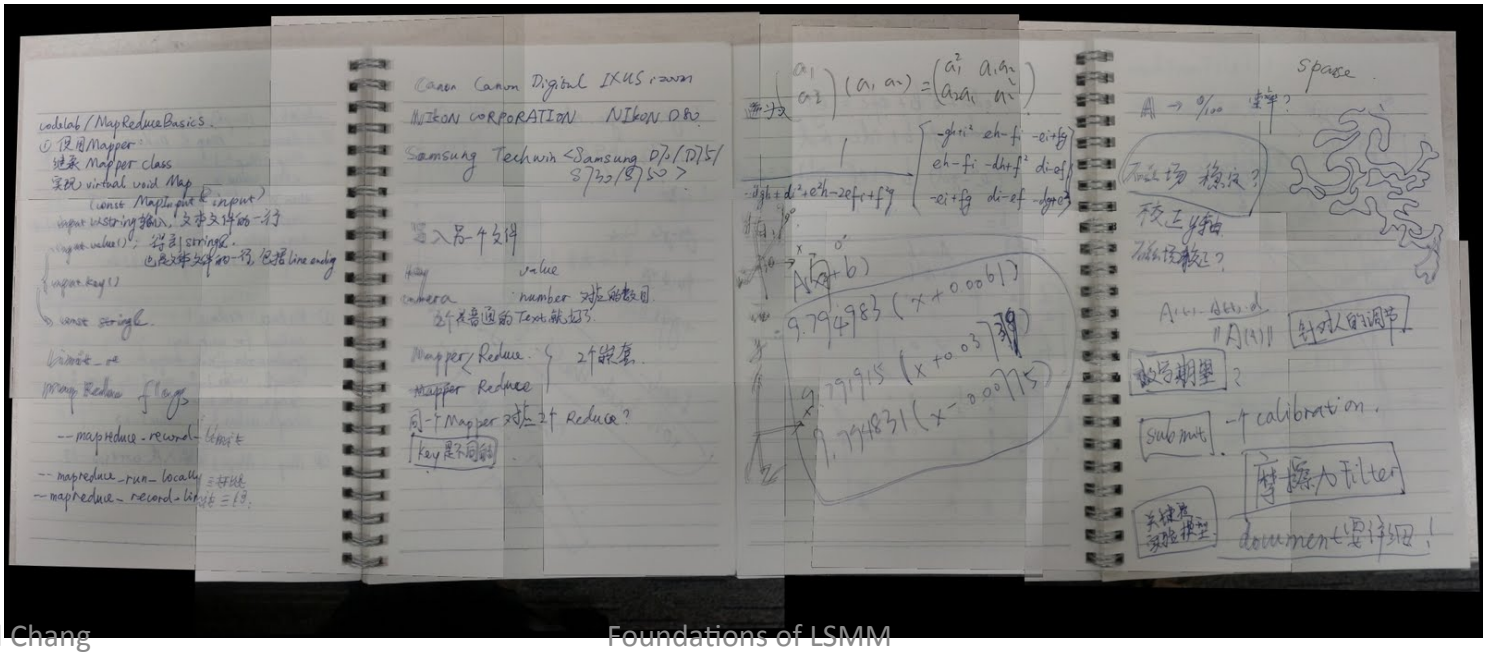


With INS



Near-Field Panoramas





More Examples…



Indoor and 3D Positioning & Navigation



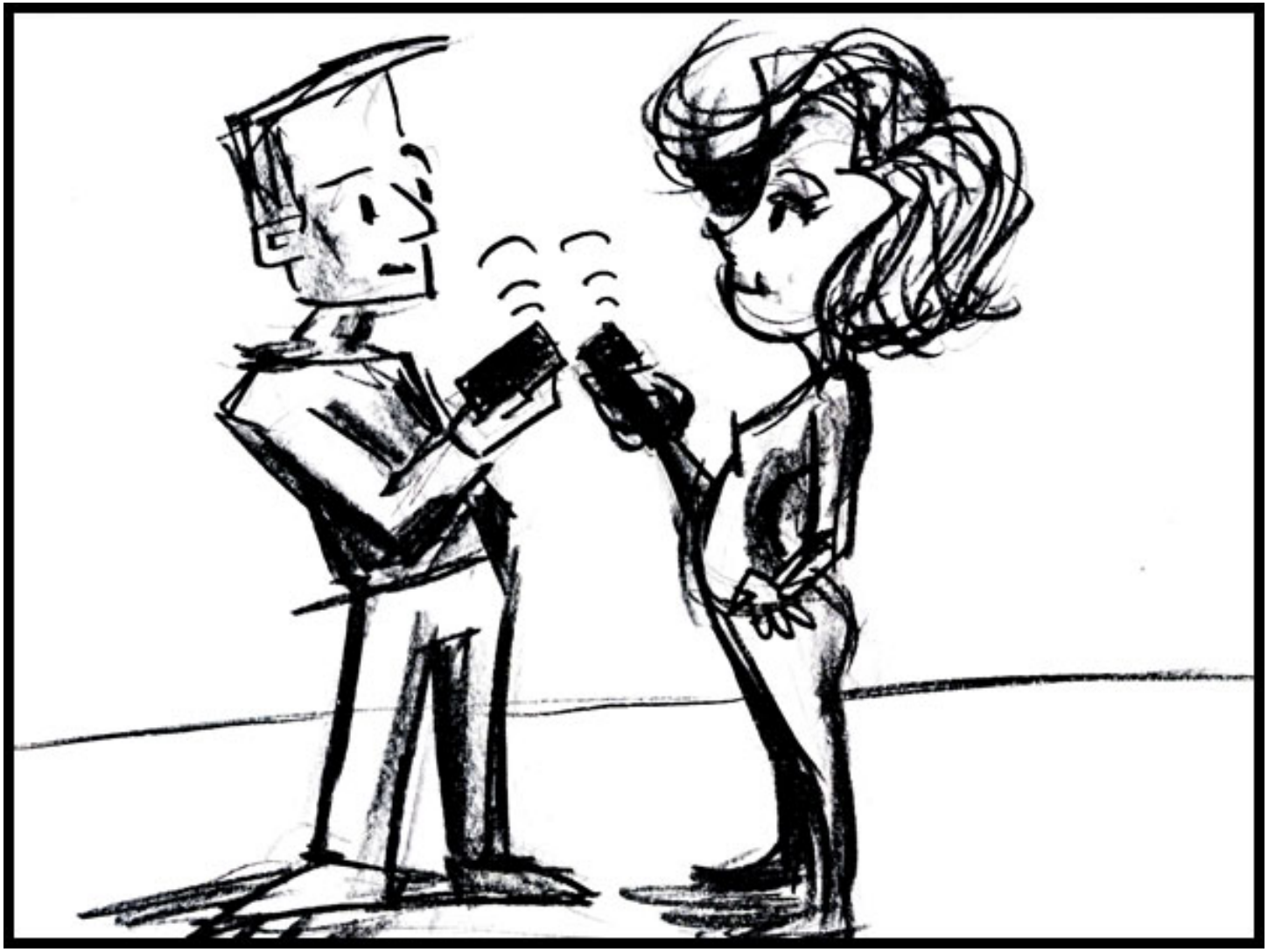
A Location-Sensitive Scenario

Alice shopping land

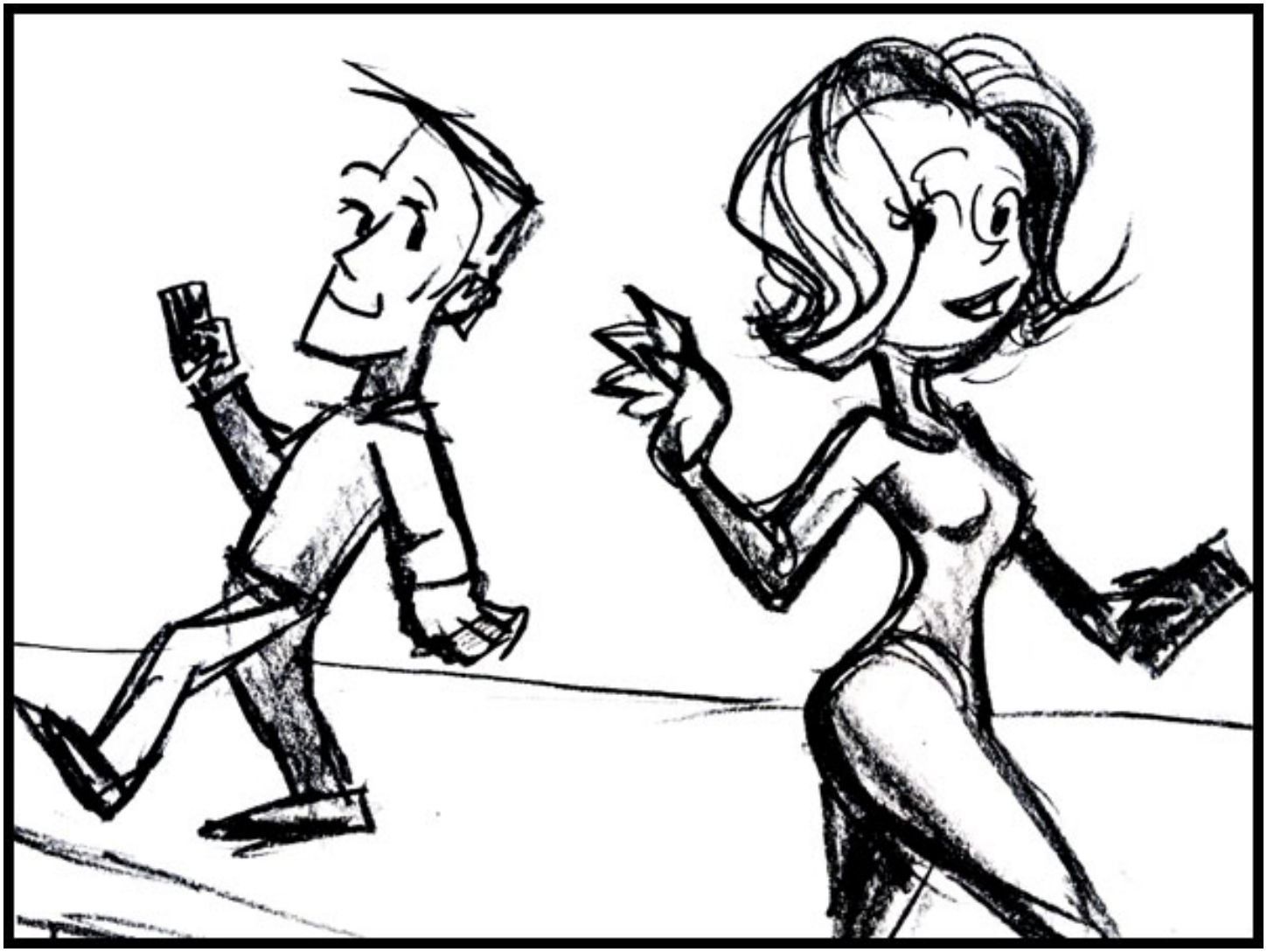


Thanks to Eric Rosenblum's drawings

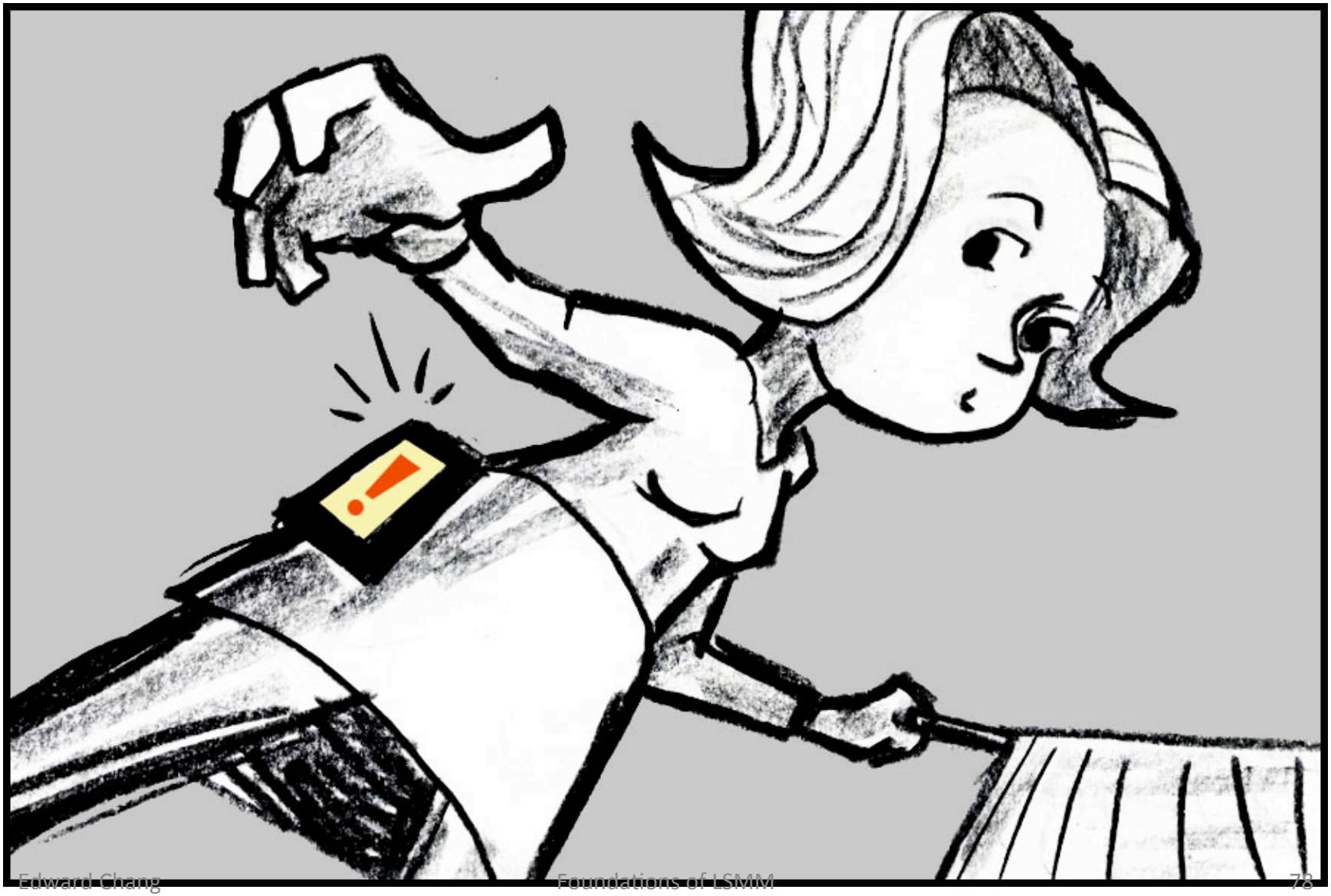
Alice: “Let us register our locations”



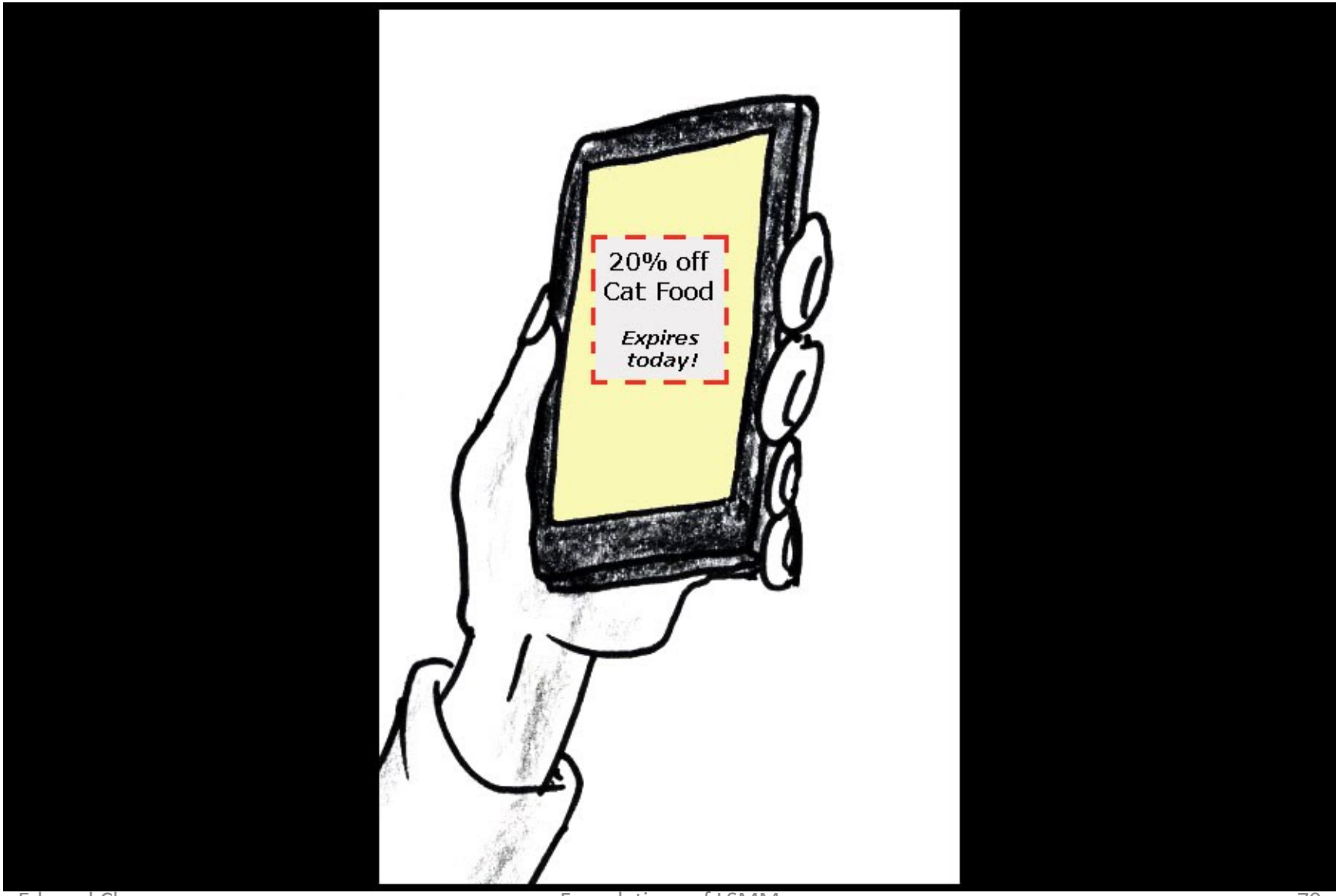
Mike: “Let me wait for you at a coffee shop”



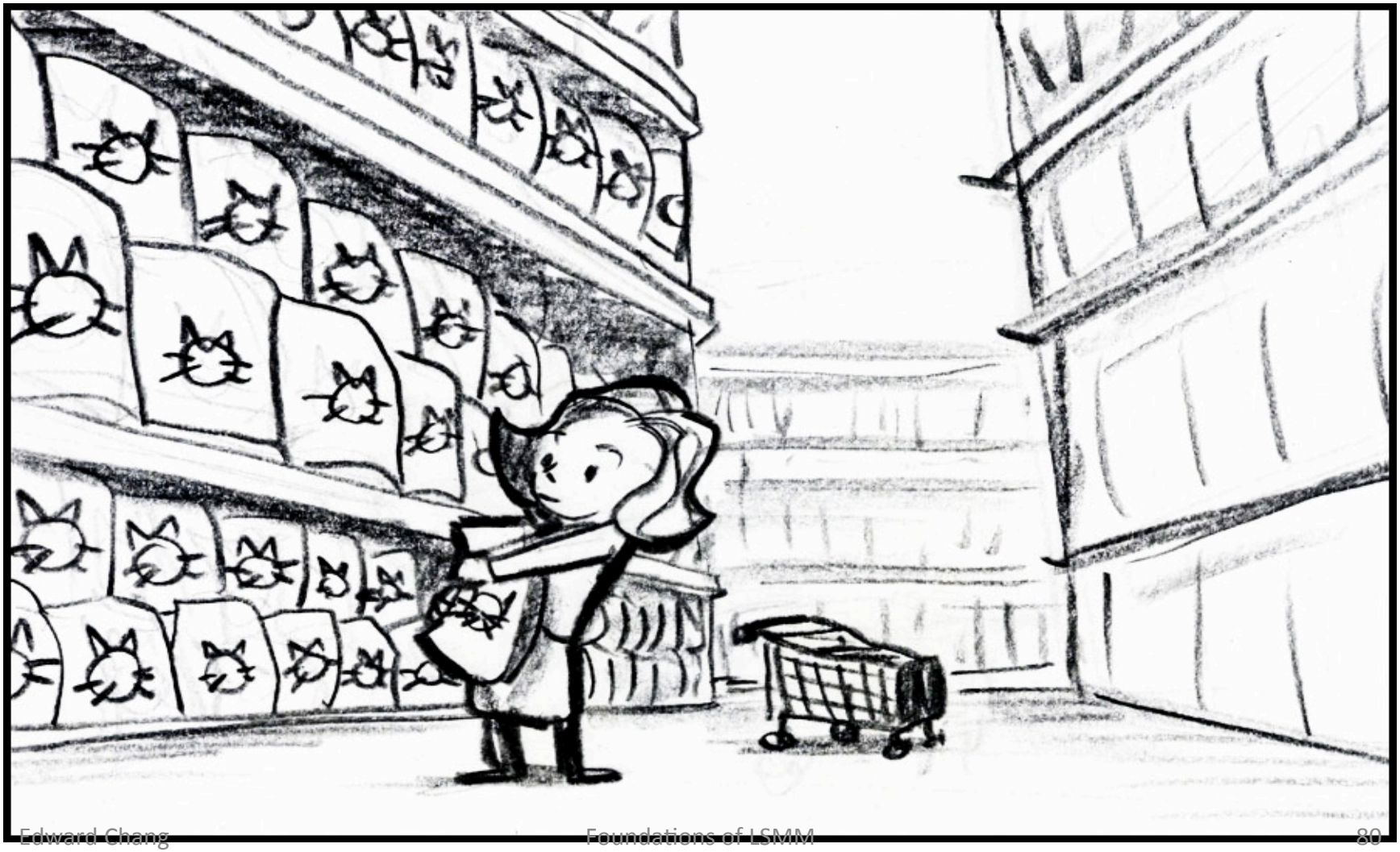
What?



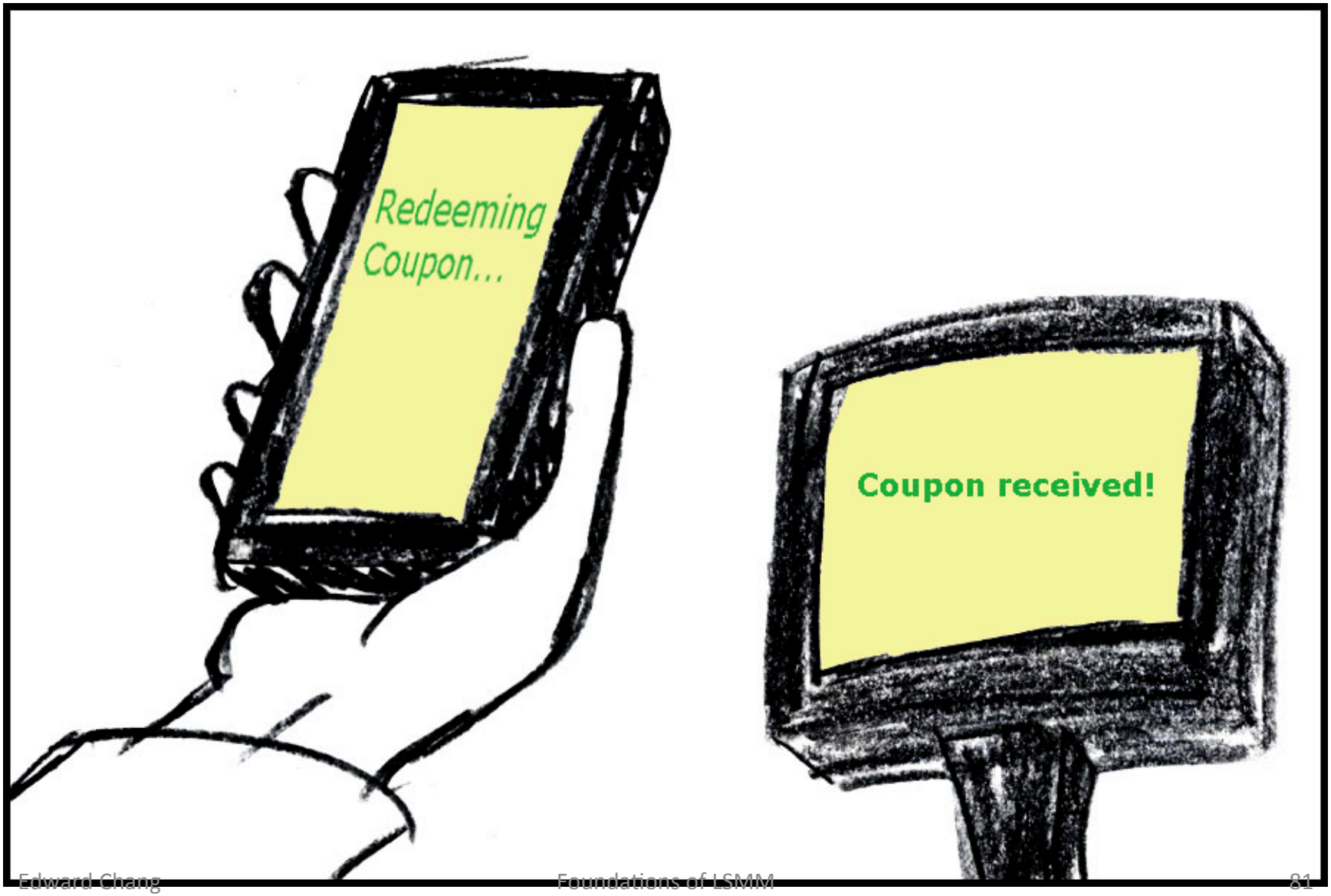
Got a relevant coupon



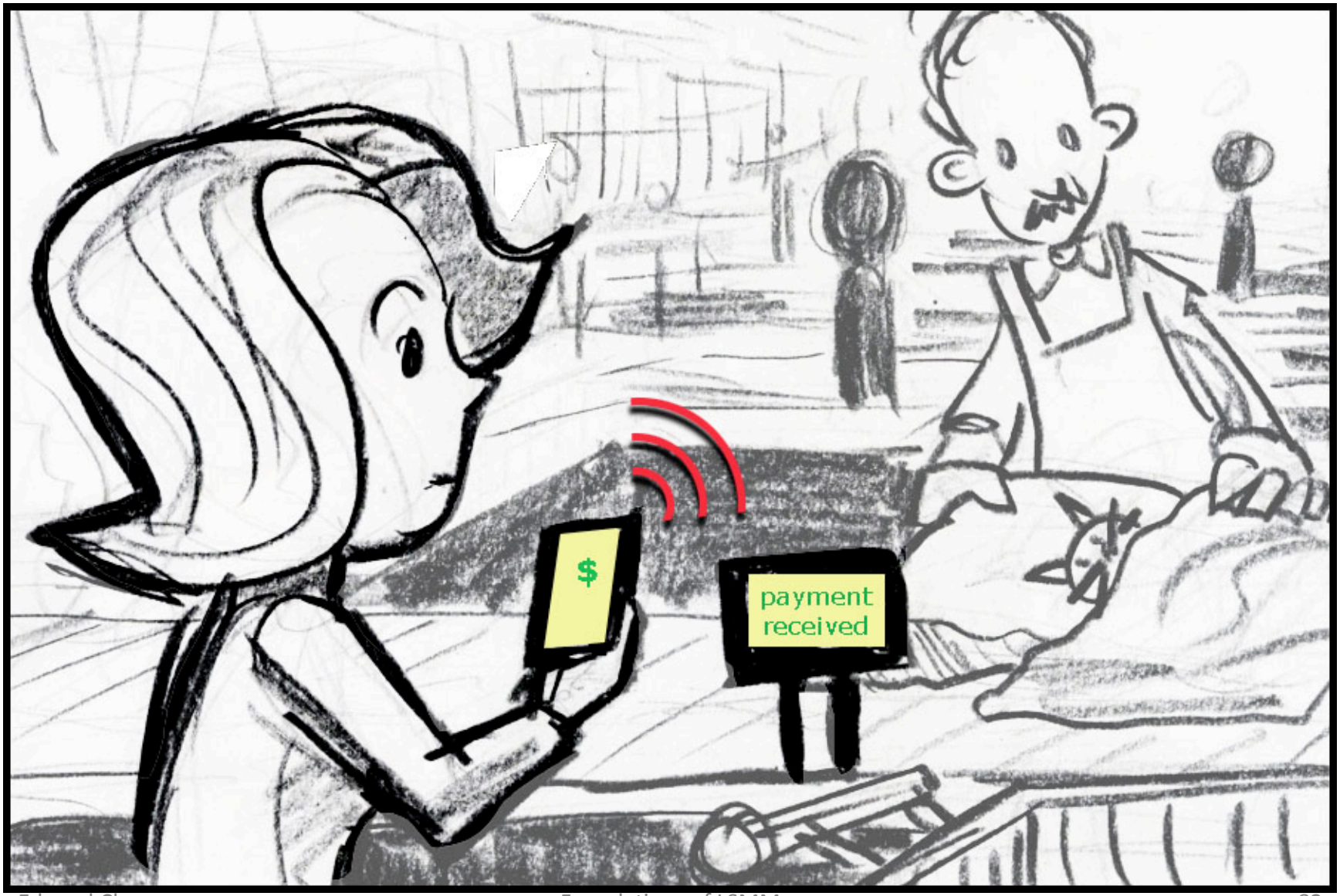
Find the store, and navigate to the cat food aisle...



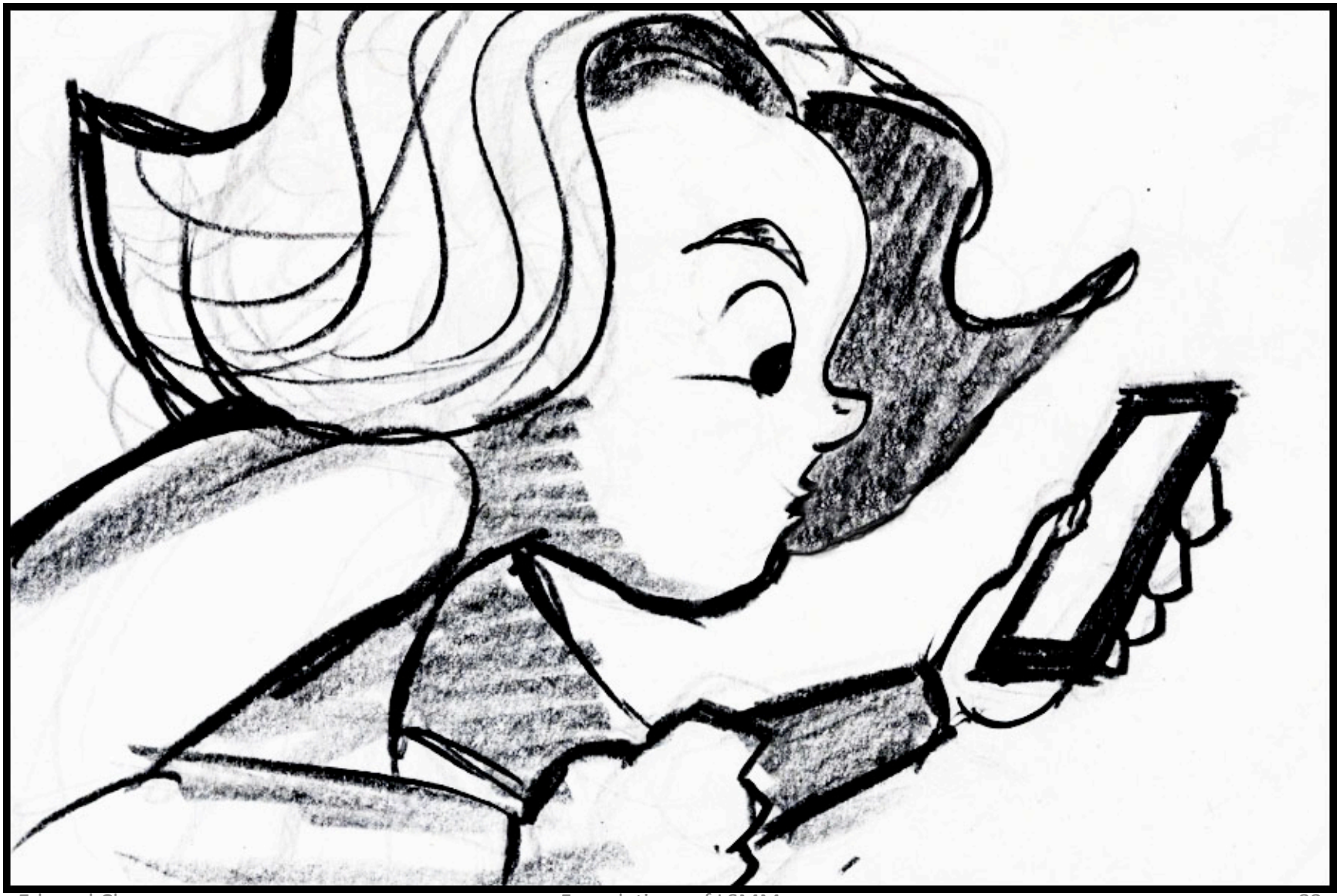
Redeem the coupon…



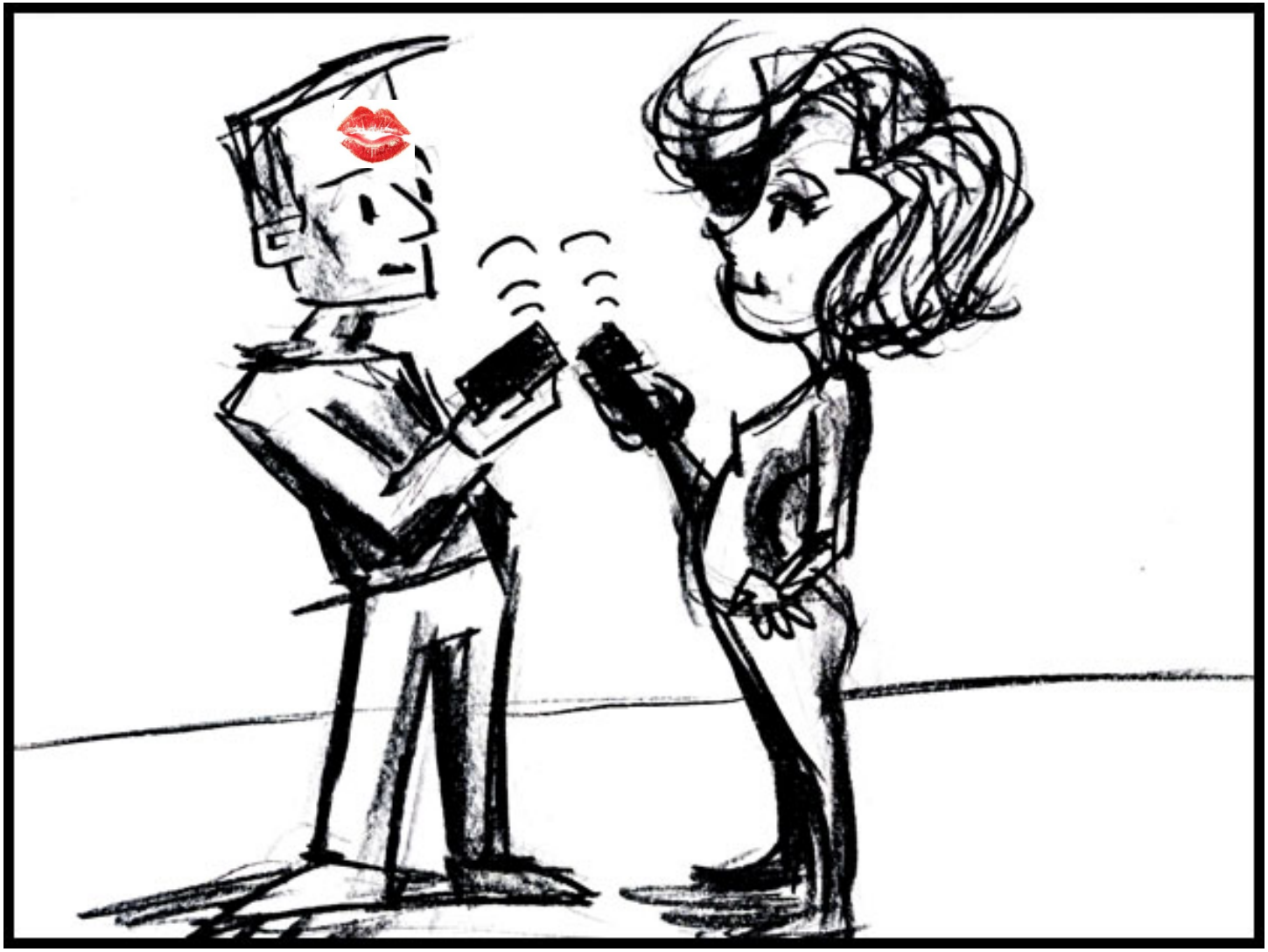
...and pay with her phone.



Looking for Mike…



…Where have you been?



Summary

- Cross-Domain Fusion
- Cross-Modality Fusion

Reading

- Foundations of Large-Scale Multimedia Information Management and Retrieval, E. Y. Chang, Springer, 2011
  - Chapter #6 Cross-Modality Fusion
  - Chapter #7 Context + Content Fusion
  - Chapter #8 Cross-Domain Fusion